

Lecture 09

Carbon Tariffs

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AEM 4510

Roadmap

1. What are carbon tariffs?
2. Basic trade theory: supply, demand, and tariffs
3. Adding pollution externalities to trade
4. Unilateral carbon taxes and leakage
5. Carbon tariffs to prevent leakage
6. Carbon tariffs alone
7. World carbon taxes
8. Different pollution intensities across countries

What are carbon tariffs?

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Carbon tariffs aim to "level the playing field" and reduce leakage

Carbon tariffs in practice

EU Carbon Border Adjustment Mechanism (CBAM):

- Phasing in 2023-2026, full implementation 2026
- Covers cement, iron/steel, aluminum, fertilizers, electricity, hydrogen
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UK CBAM: announced for 2027

US proposals: various bills proposed but none passed yet

Understanding border carbon adjustments



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The key insight: carbon tariffs are equivalent to a **consumption tax** plus a **production subsidy**

Basic trade theory

Tariffs are taxes on imports

A tariff is a **tax on imports**

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In this lecture we study the **small-country** case:

- country i is too small to affect the world price
- imports are available at the exogenous world price P_W
- this is the benchmark that makes leakage especially transparent later on

A simple trade model

Consider a small open economy (country i) that:

- Takes world prices as given
- Can import or export a homogeneous good

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Key curves:

- S : domestic supply
- D : domestic demand
- P_W : world price
 - The social marginal benefit of the good

Domestic suppliers can't charge more than P_W , and won't charge less

Supply and demand refresher

In this market:

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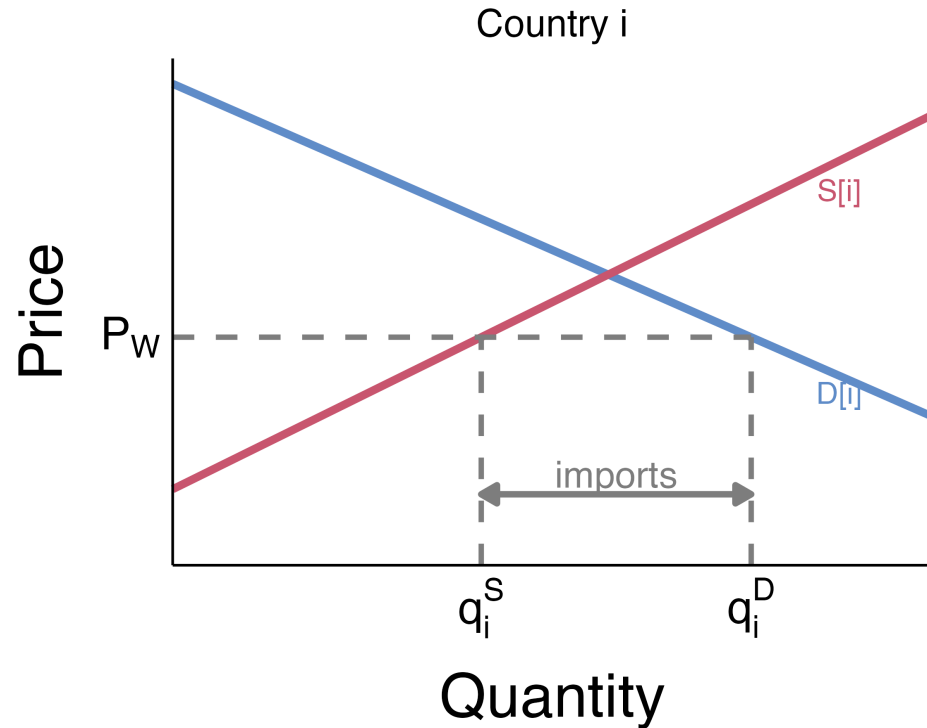
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- The gap between domestic demand and domestic supply is imports

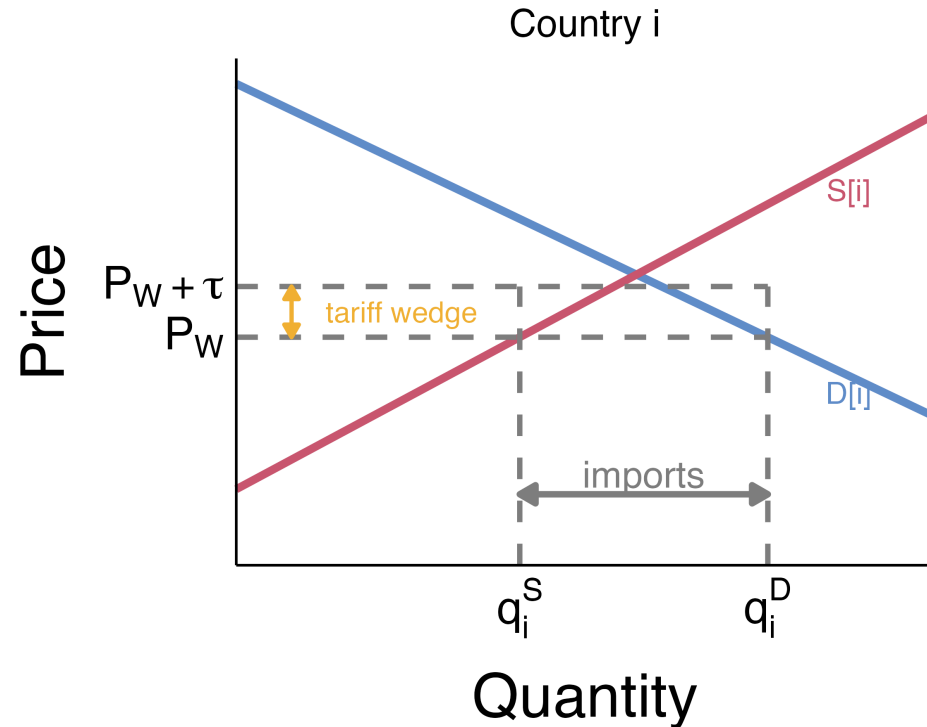
Free trade benchmark



With free trade at price P_W :

- Domestic production is q_i^S
- Domestic consumption is q_i^D
- Imports are $q_i^D - q_i^S$

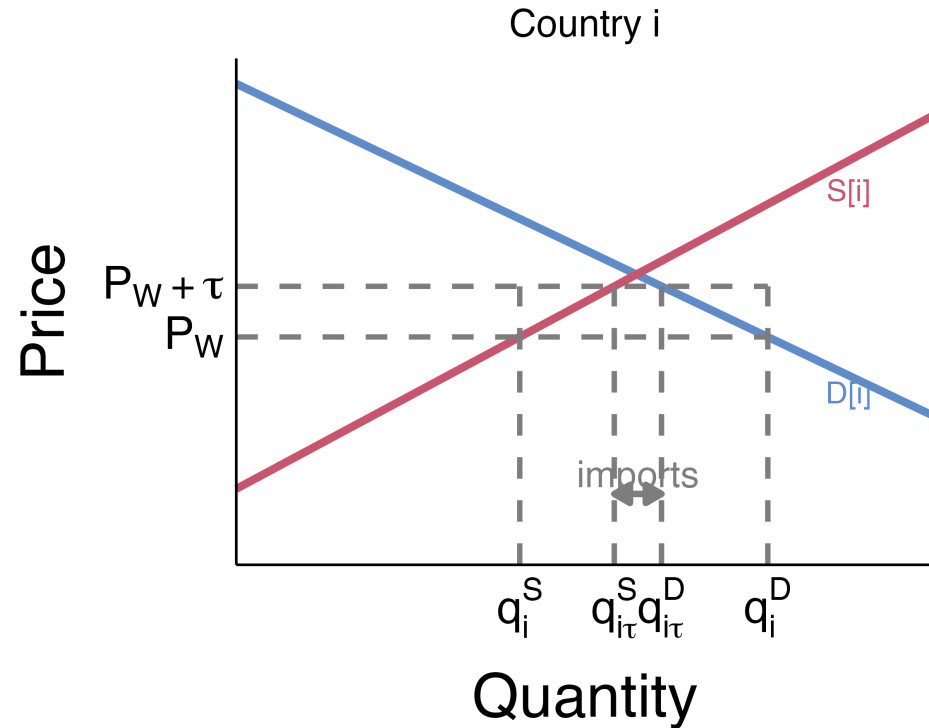
The tariff wedge



Suppose we implement an import tariff τ :

- The domestic price rises from P_W to $P_W + \tau$
- Imported units now enter at the tariff-inclusive price
- Domestic suppliers can charge that same higher price

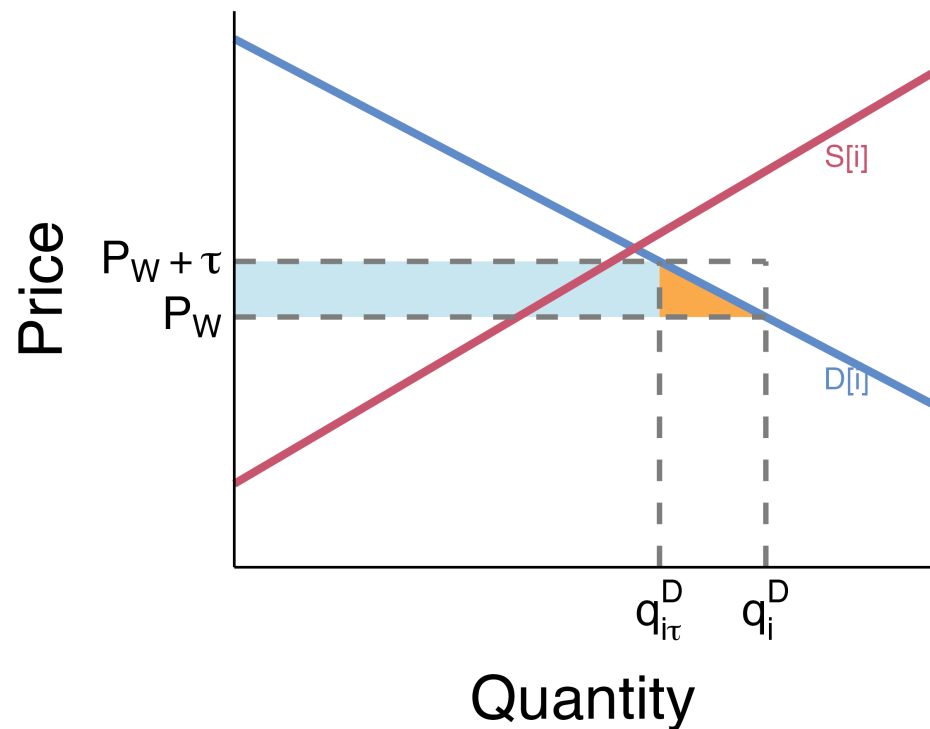
Quantity effects of a tariff



A tariff τ raises the domestic price to $P_W + \tau$:

- Domestic production rises to $q_{i\tau}^S$
- Domestic consumption falls to $q_{i\tau}^D$
- Imports shrink to $q_{i\tau}^D - q_{i\tau}^S$

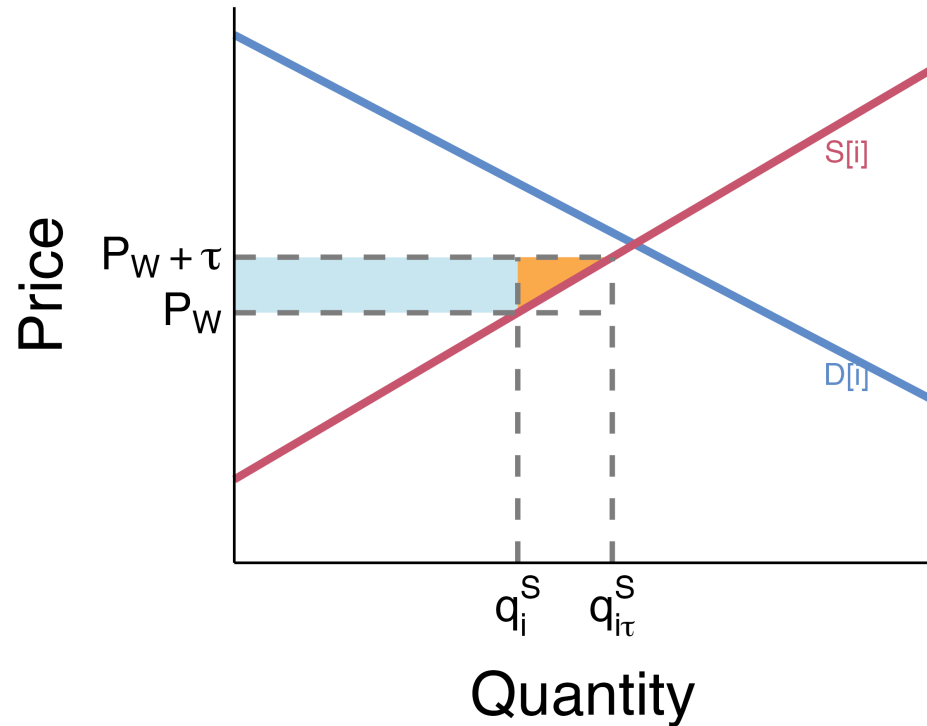
Consumer surplus refresher



Consumer surplus is the area below demand and above price.

- Blue rectangle: consumers pay more on the units they still buy
- Orange triangle: consumers stop buying units they still valued above P_W

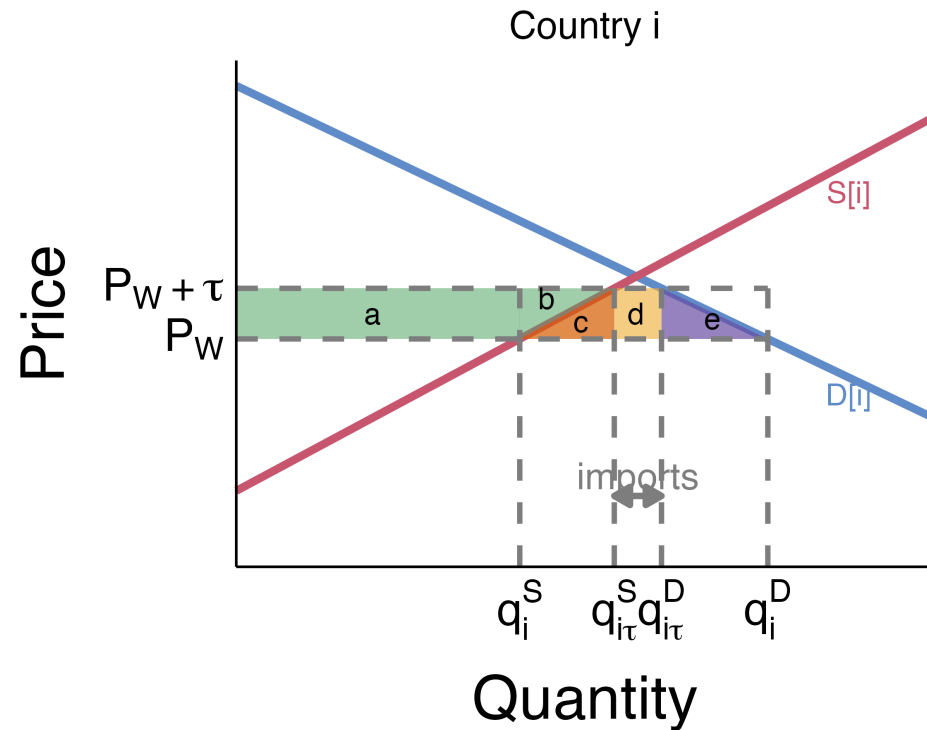
Producer surplus refresher



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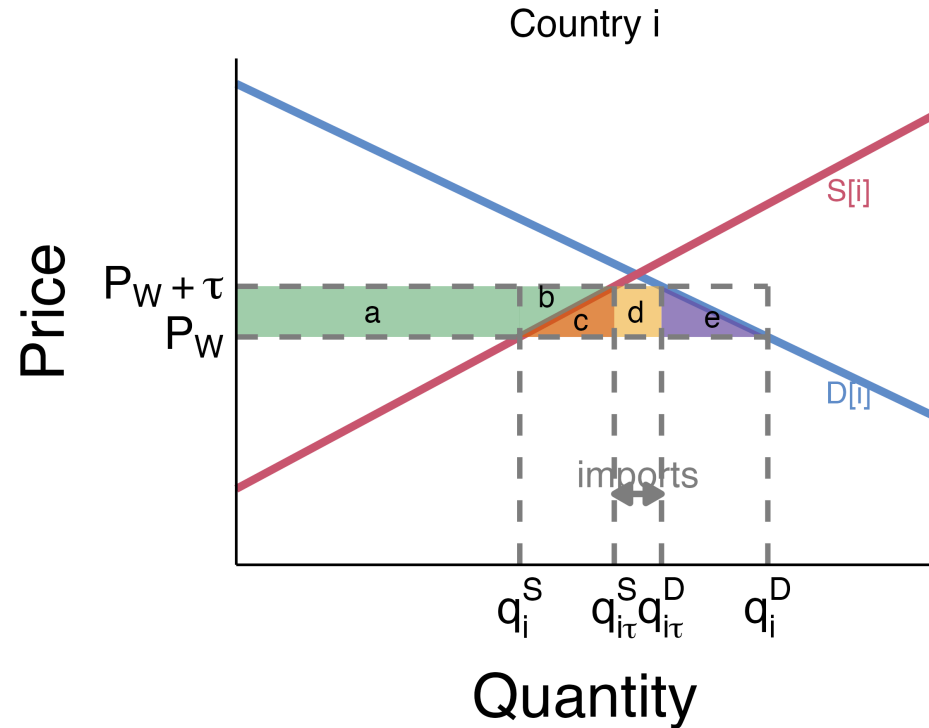
- Blue rectangle: producers receive a higher price on their original output
- Orange triangle: producers gain extra surplus when they expand output to $q_{i\tau}^S$

Welfare accounting under a tariff



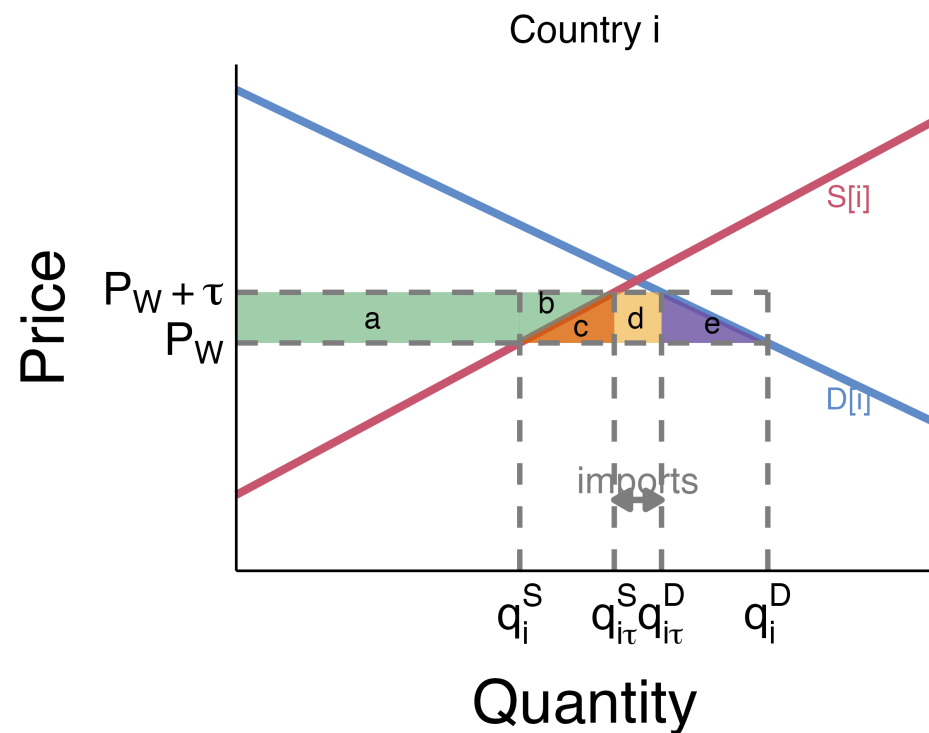
- a : producer gain on the original domestic output
- b : producer gain on the extra domestic output
- c : production deadweight loss
- d : tariff revenue on remaining imports
- e : consumption deadweight loss

Consumer loss and redistribution



- $-\Delta CS = a + b + c + d + e$
- The whole shaded area is the **consumer surplus loss**
- But $a + b$ reappear as part of **producer surplus gain**
- And d reappears as **tariff revenue**

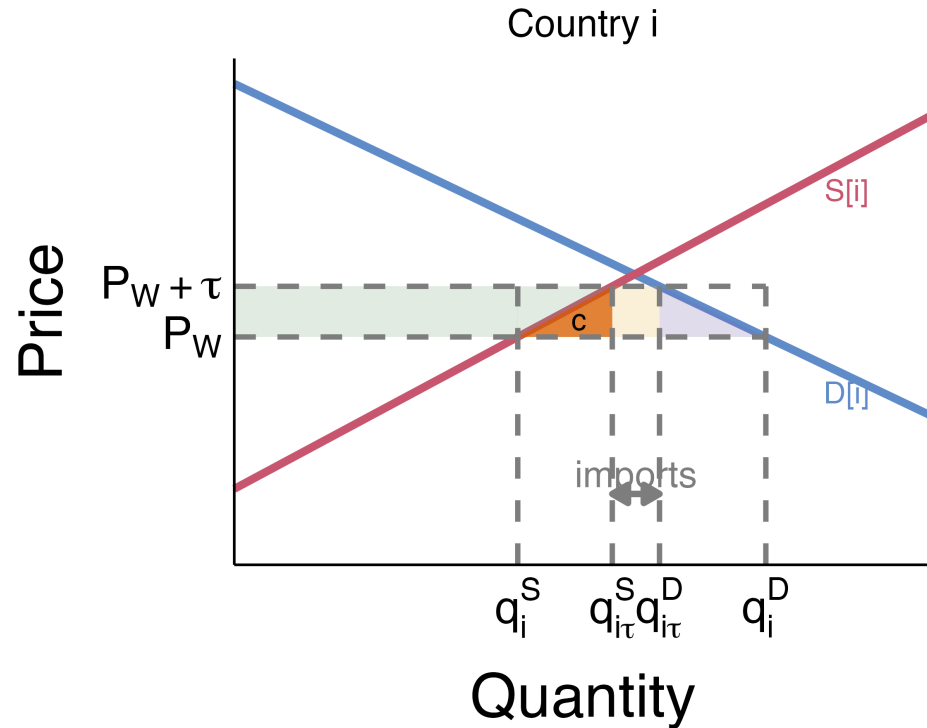
Tariff welfare accounting



- $\Delta PS = a + b$
- $\Delta CS = -(a + b + c + d + e)$
- $R = d$
- $\Delta W_i = \Delta PS + R - \Delta CS$
- So for a **small country**,

$$\Delta W_i = -c - e$$
- The only net welfare losses are the two deadweight-loss regions c and e

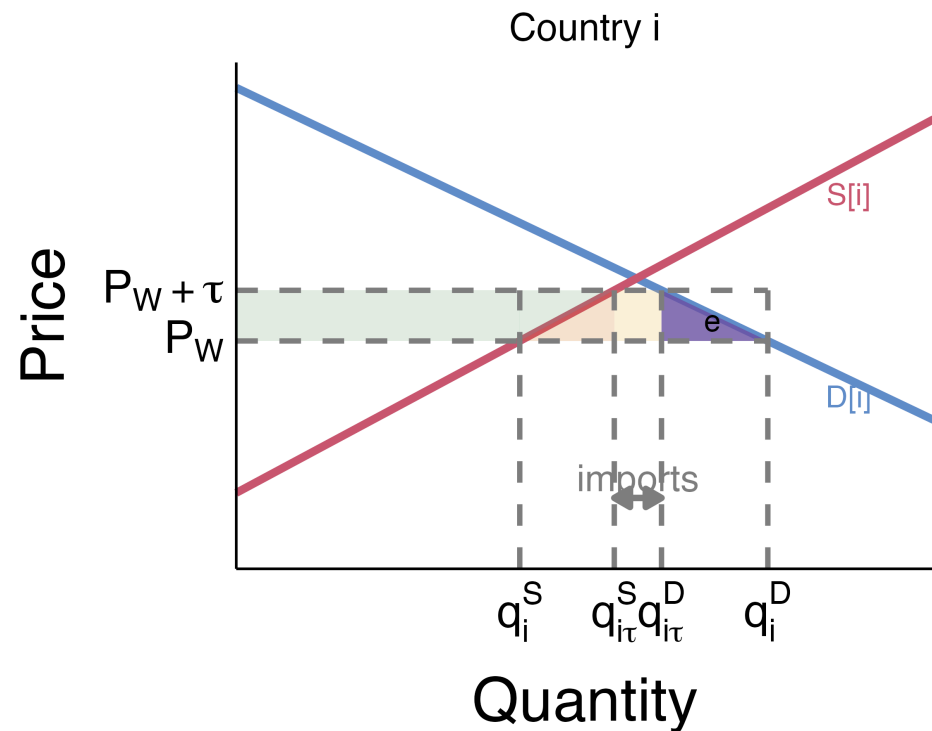
Production distortion



Triangle c is a production distortion:

- The tariff shifts production from lower-cost foreign supply to higher-cost domestic production
- For those extra domestic units, $MC (S[i])$ exceeds $MB (P_W)$

Consumption distortion



Triangle e is a consumption distortion:

- The tariff eliminates units consumers value above the world price but below the tariff-inclusive price
- For those lost units, MB ($D[i]$) exceeds MC (P_W)

Why start with ordinary tariffs?

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So next we add the environmental externality to this standard tariff benchmark

Adding pollution

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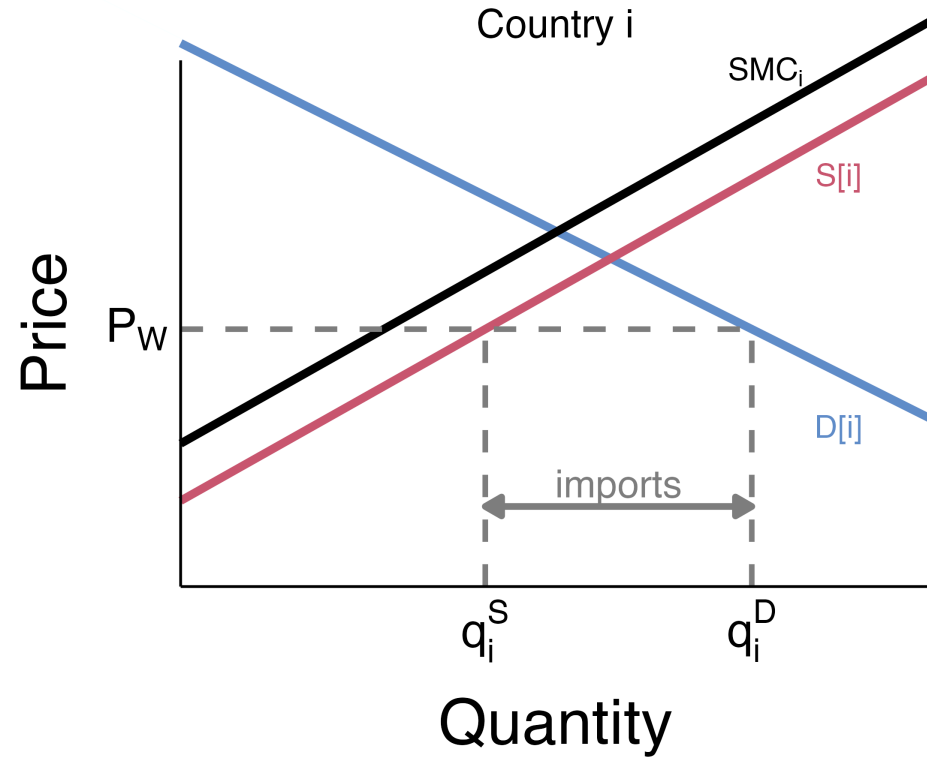
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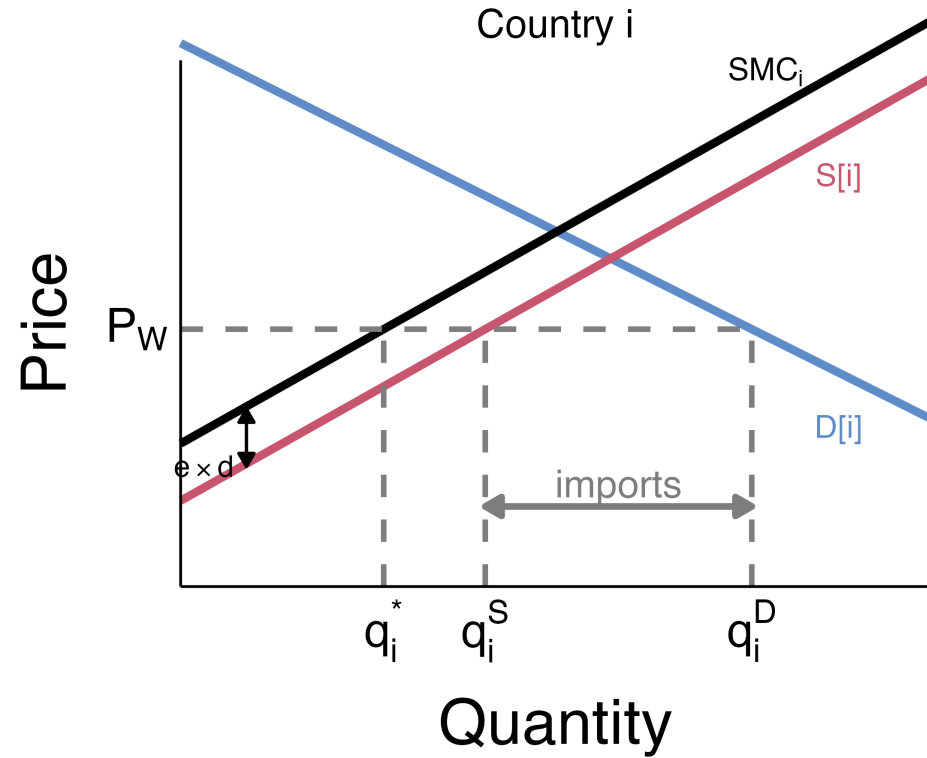
SMC lies above the private supply curve by the external cost $e \times d$

Free trade with pollution



Free trade output is pinned down by domestic demand and supply at the world price

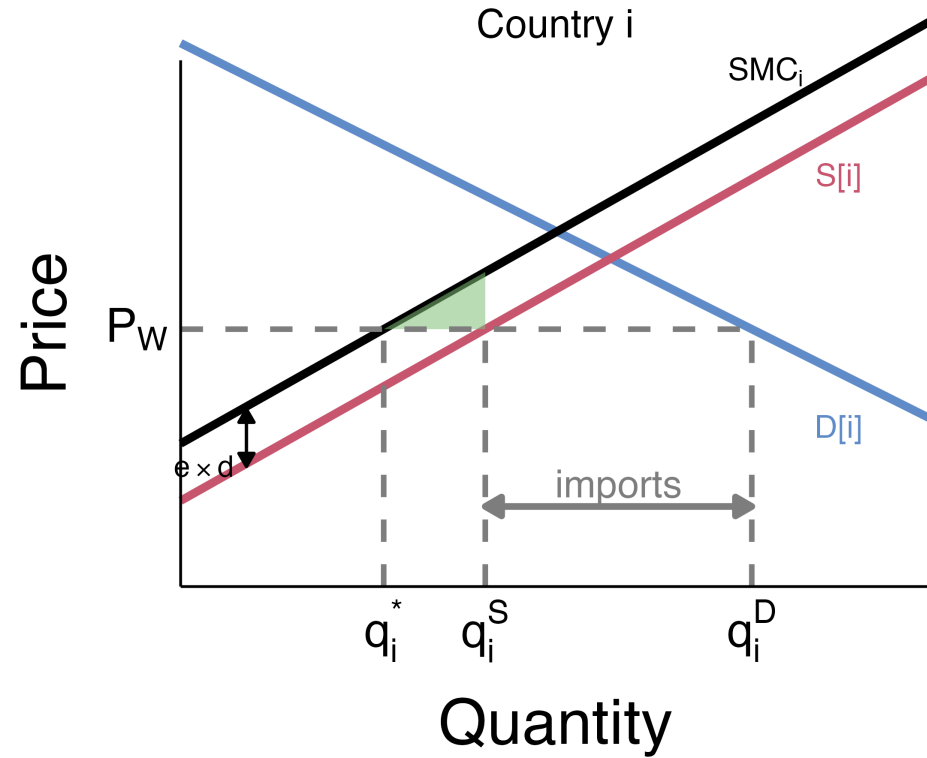
Free trade with pollution



Country i 's SMC curve lies above its private supply curve

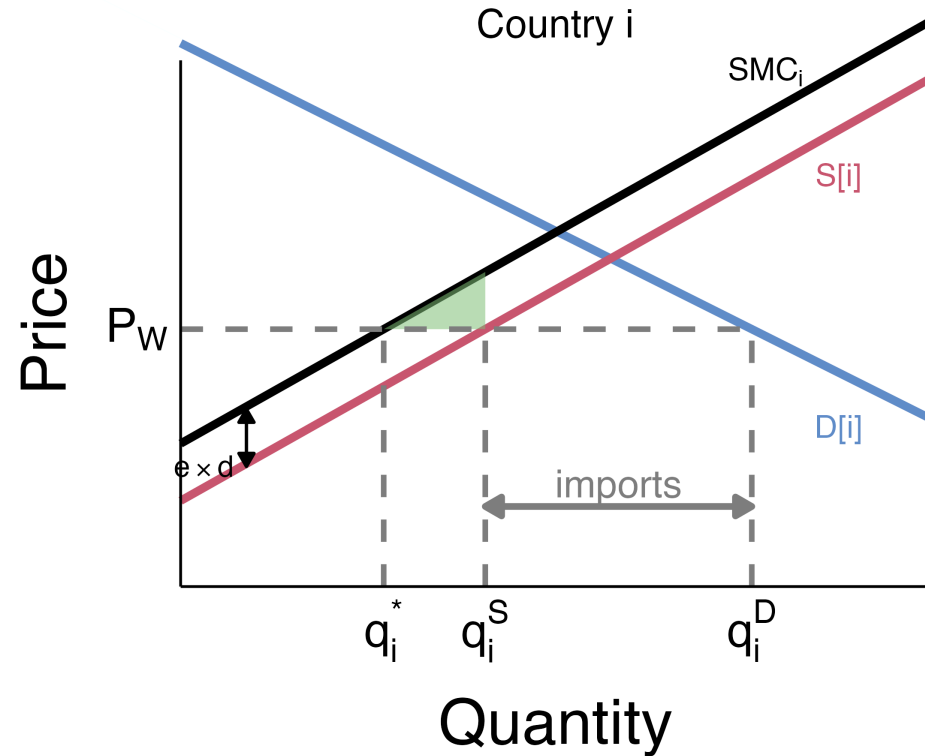
The vertical gap is the external cost: $e \times d$

Free trade with pollution



The green region is the domestic deadweight loss from producing beyond the efficient output level

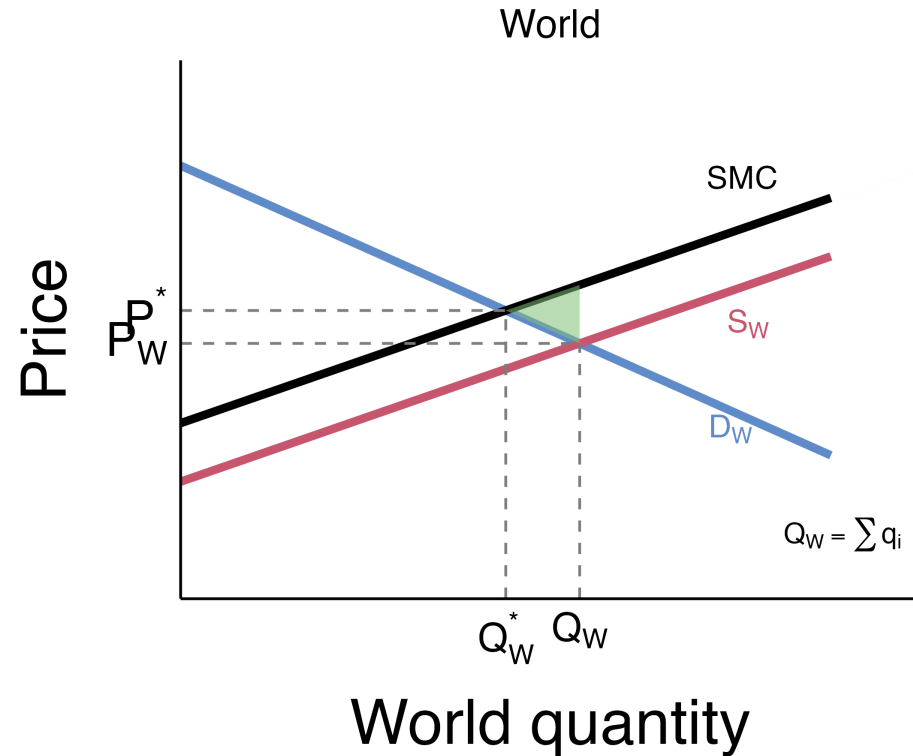
Free trade with pollution



At the world price P_W , marginal benefit is pinned down by world demand, while marginal social cost is given by SMC

That triangle is the set of units for which marginal benefit exceeds private supply cost but is below marginal social cost

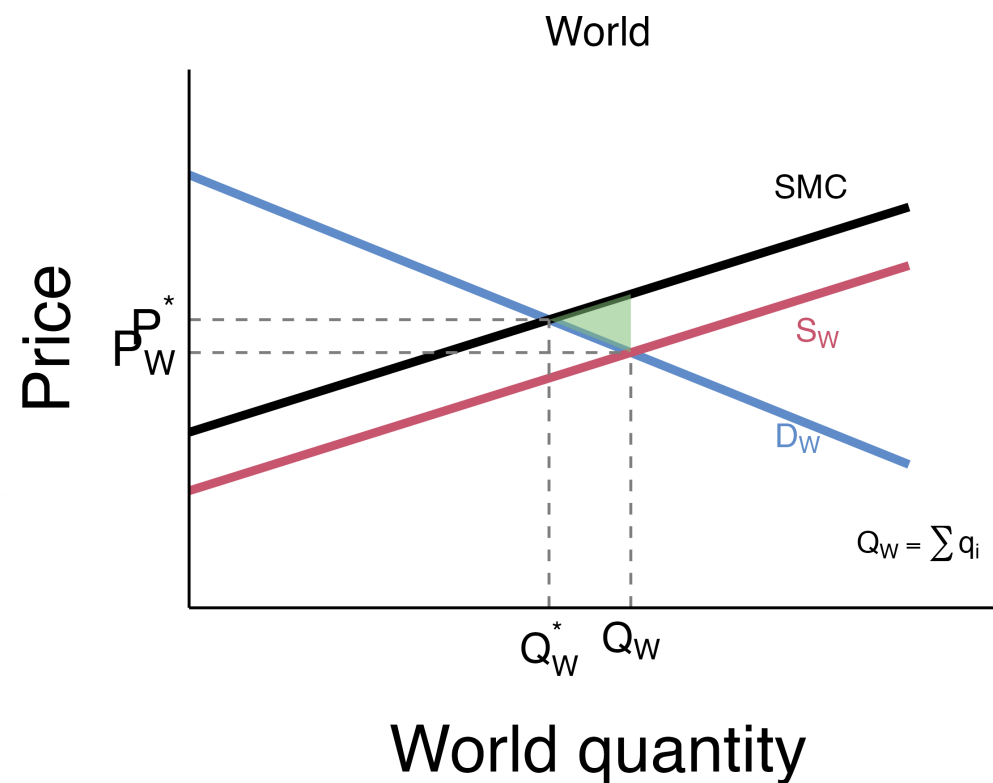
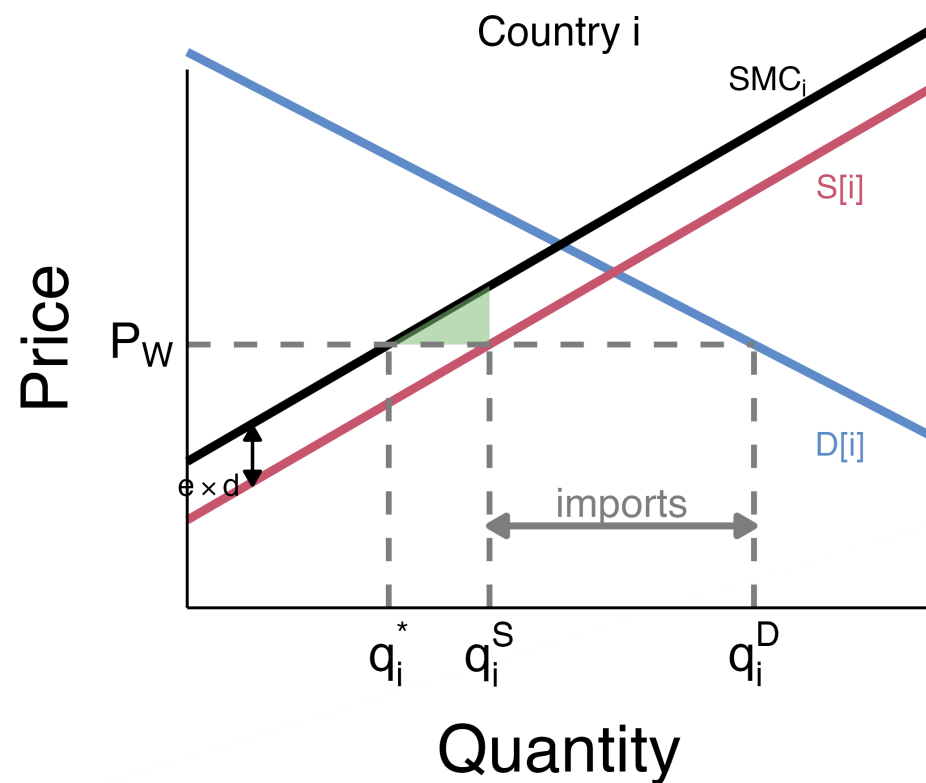
Global perspective



From a global perspective:

- Free-trade output is where $D = S$
- Efficient output is where $D = SMC$
- The green region is the global welfare gain from cutting output from Q_W to Q_W^*

Country and global perspectives



Global DWL on the right is the sum of each country's DWL on the left

The externality problem

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But what if only **one country** acts?

Unilateral carbon tax

Country i imposes a carbon tax

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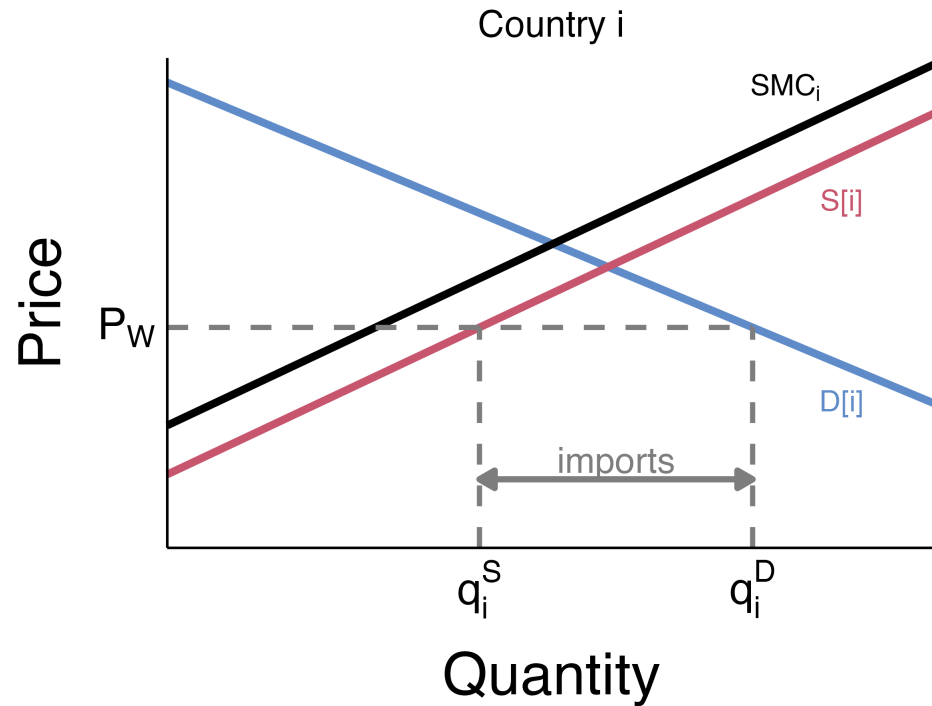
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So the carbon tax reduces the net price domestic producers receive from P_W to $P_W - t$

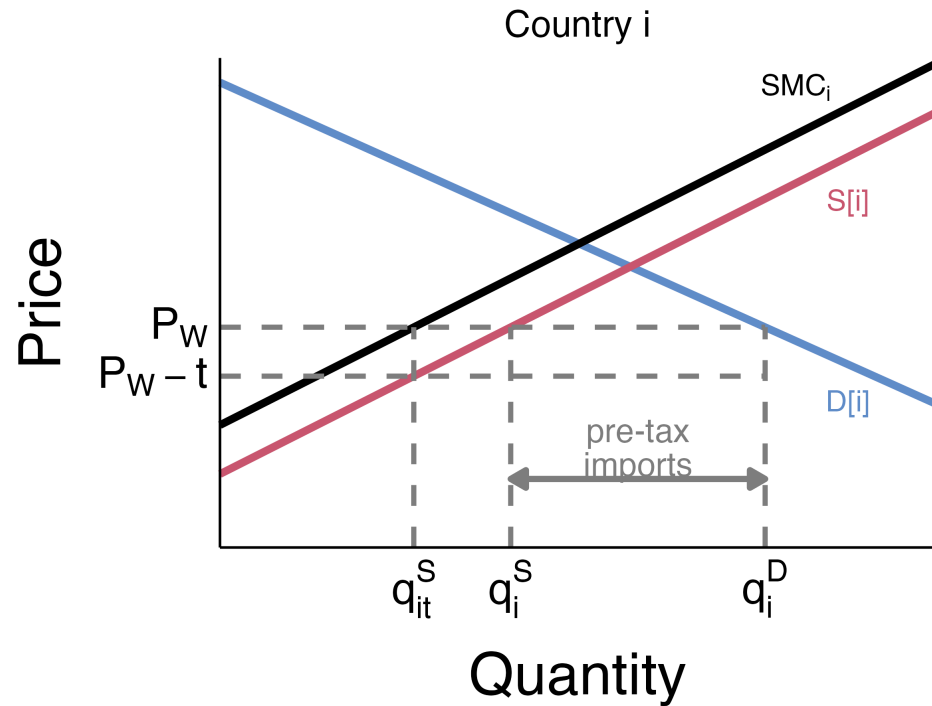
Effects of unilateral carbon tax



Before the tax, the domestic market is at the free-trade benchmark with price P_W

- You can think of a tax $t = e \times d$ as shifting $S[i]$ up to $SMC[i]$, so firms produce until $S[i] + t = P_W$
- Equivalently, firms keep the same supply curve $S[i]$ but face the net-of-tax price $P_W - t$

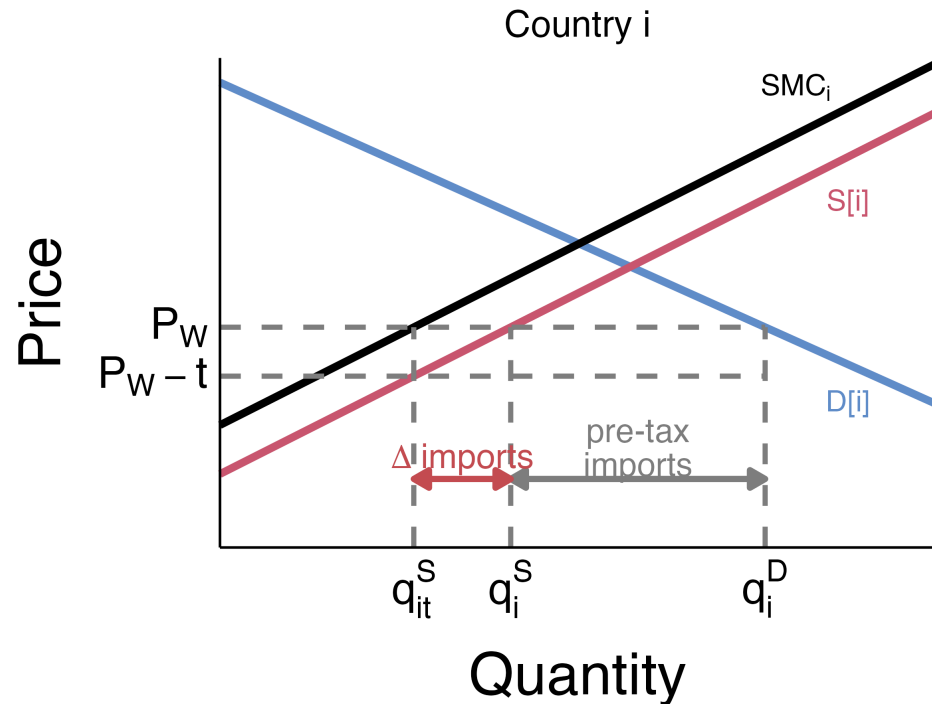
Effects of unilateral carbon tax



After the tax:

- Consumers still face the world price P_W
- But with a tax t , producers receive the net-of-tax price $P_W - t$
- Domestic output falls to q_{it}^S

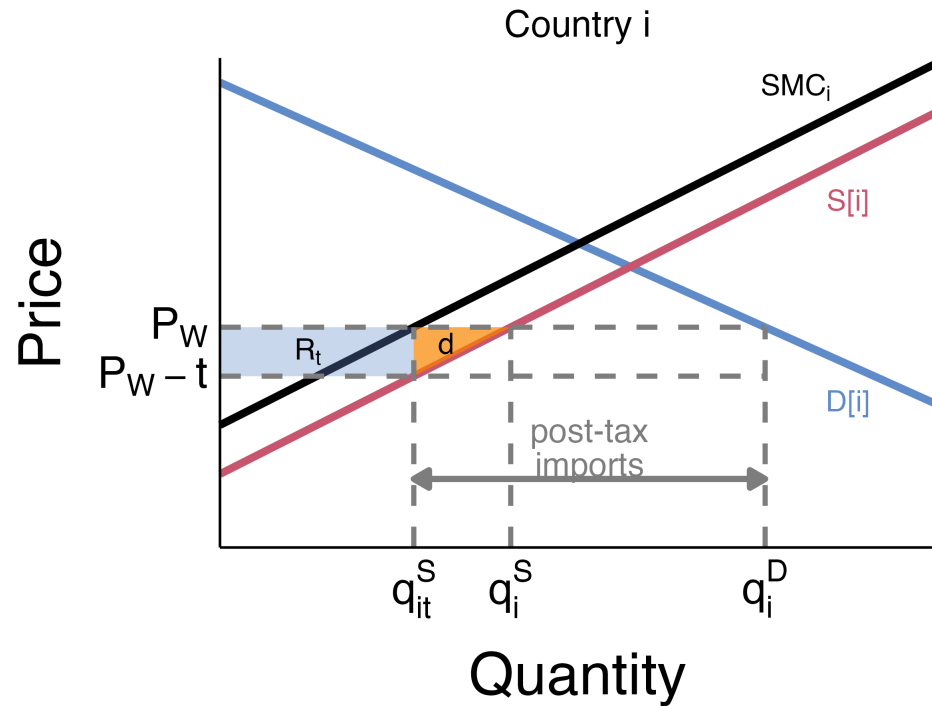
Effects of unilateral carbon tax



Leakage comes next:

- Consumers in country i still demand q_i^D at price P_W
- Imports rise enough to offset the fall in domestic output
- Foreign production replaces home production: that is leakage
- **Global** quantity produced (and emissions) does not change

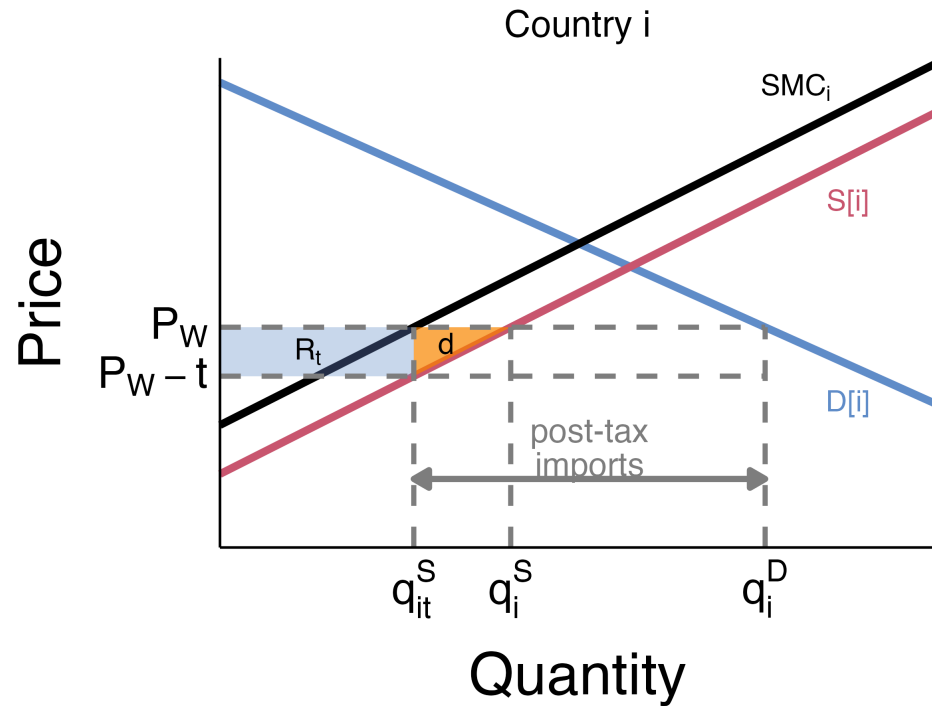
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From country i 's perspective:

- Consumer surplus does not change: they still pay P_W and consume q_i^D
- The government gains revenue R_t

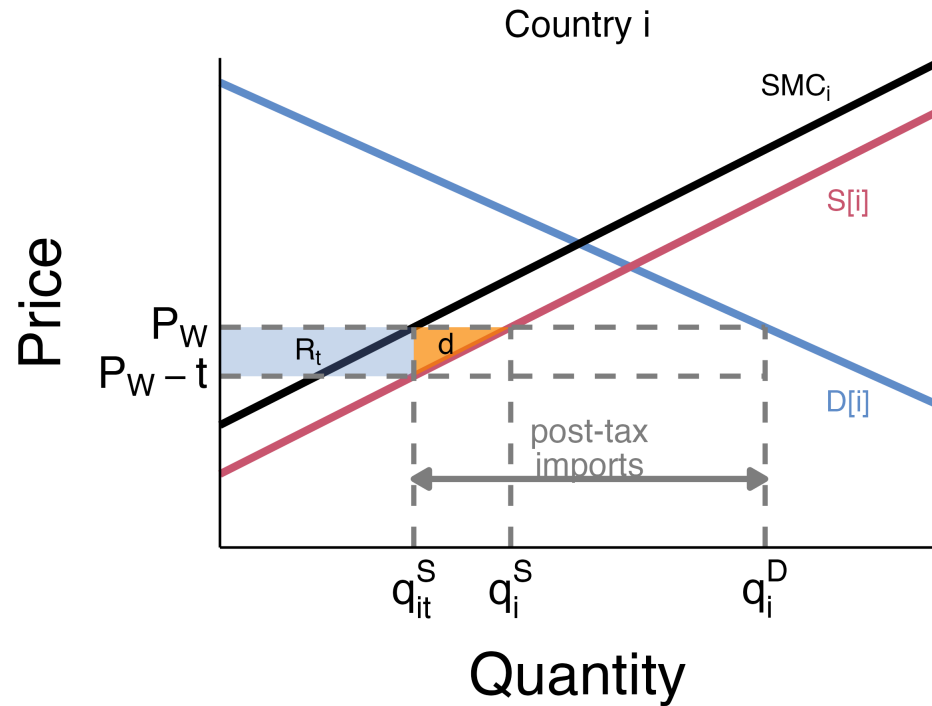
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From country i 's perspective:

- Producers lose producer surplus equal to $R_t + d$

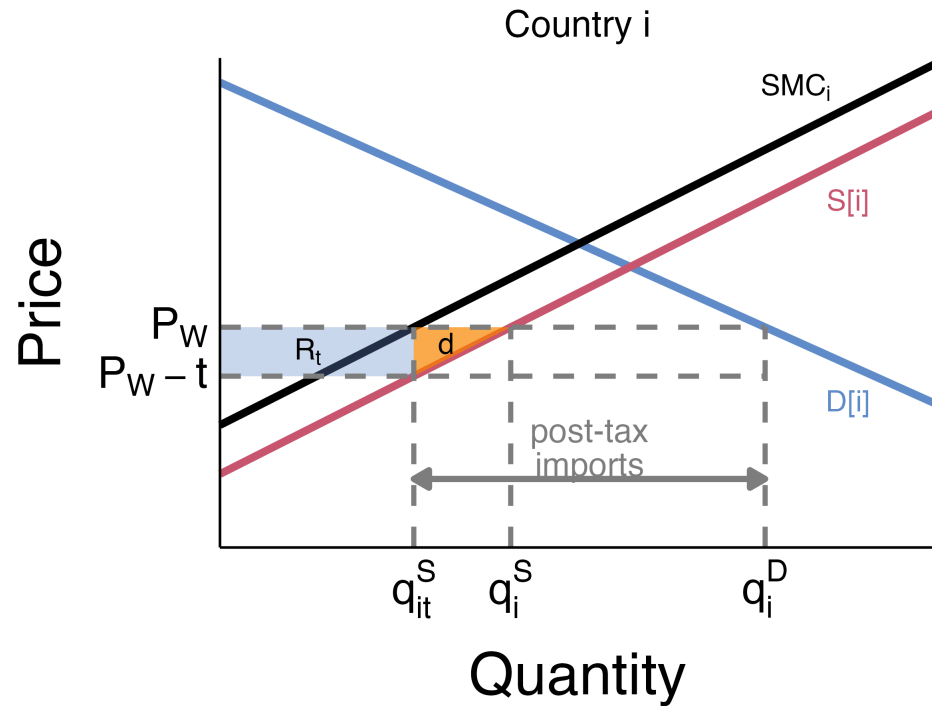
Effects of unilateral carbon tax



Why do producers lose $R_t + d$?

- Blue rectangle: they receive a lower net-of-tax price on the units they still sell
- Orange triangle: they also stop selling the units between q_{it}^S and q_i^S for which their $PMC < P_W$

Effects of unilateral carbon tax

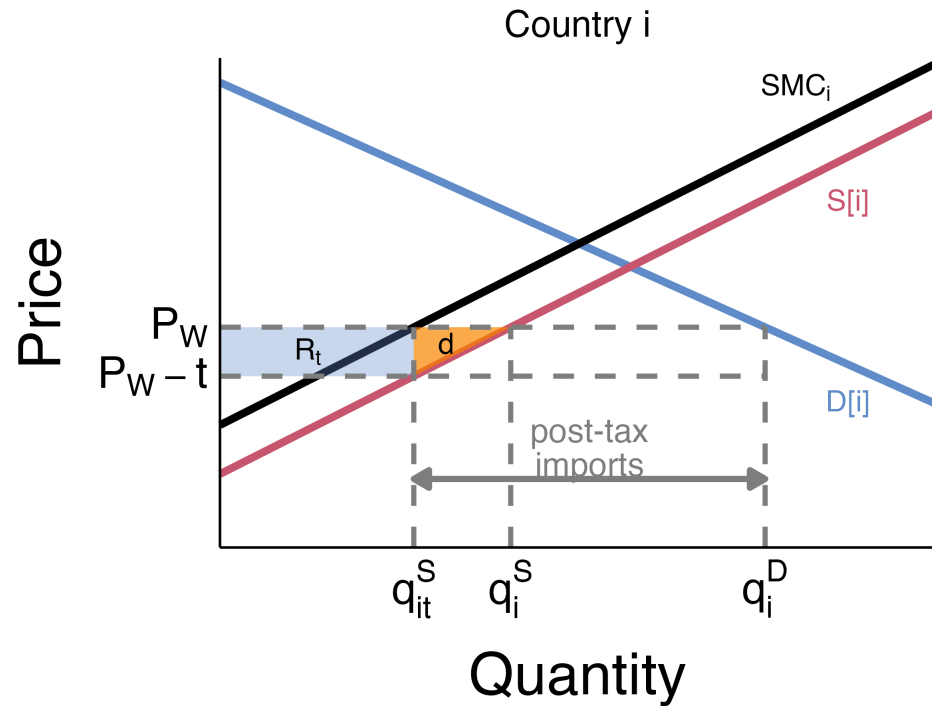


Why is d the country's deadweight loss?

- R_t is only a transfer from producers to the government, so it stays inside country i
- Net national effect:

$$+R_t - (R_t + d) = -d$$

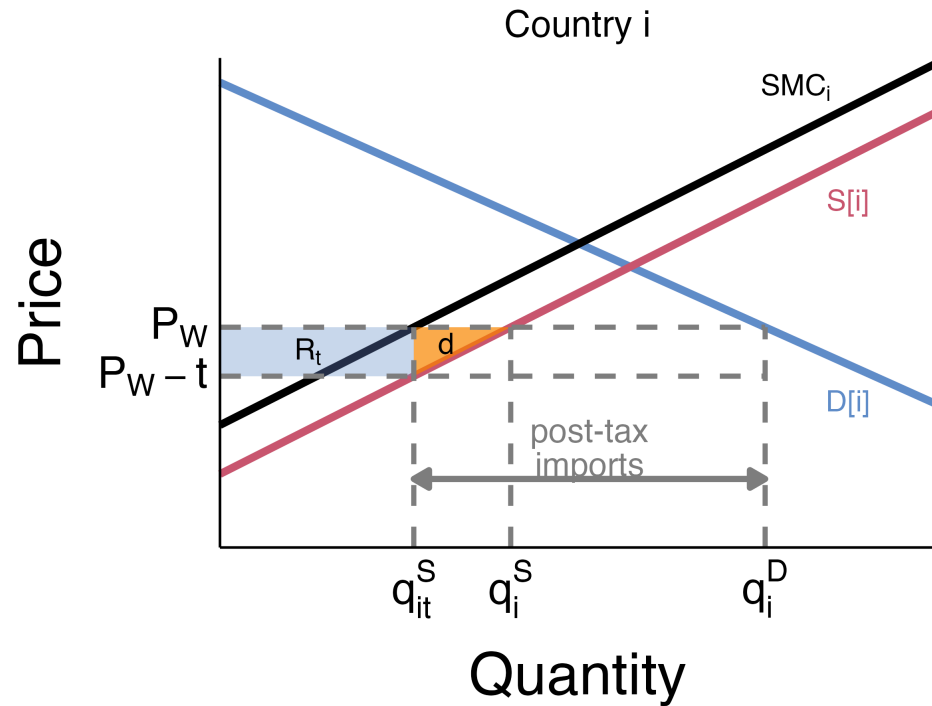
Effects of unilateral carbon tax



Why does leakage make d the relevant national loss?

- The triangle d is the pure private loss left after netting out the transfer R_t
- Consumers in country i are unaffected and imports replace the lost domestic output/emissions

Effects of unilateral carbon tax



Why does leakage make d the relevant national loss?

- The tax reduces home production, but foreign production rises 1-for-1: that is leakage
- The environmental gain is offset abroad, so country i is left with the private DWL d

Unilateral carbon tax effects

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- World price is unchanged at P_W so quantity demanded domestically for the good does not change
- Imports from abroad fill in the difference
- Production and emissions increase abroad

Leakage undermines the tax

With identical carbon intensities and a small country implementing the tax (so P_W is fixed):

- 1 unit fewer domestic production = 1 unit more foreign production
- **Net global emissions unchanged!**

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The carbon tax:

- Generates tax revenue R_t (rectangle)
- Creates deadweight loss d (triangle) from production distortion
- Achieves **zero environmental benefit**

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Country i bears costs but gets no environmental gain

Leakage is a rate, not a constant

Economists often summarize leakage with the **leakage rate**:

$$\text{Leakage rate} = \frac{\Delta E_j}{-\Delta E_i}$$

where $\Delta E_i < 0$ is the change in home emissions and $\Delta E_j > 0$ is the change in foreign emissions

Leakage is a rate, not a constant

$$\text{Leakage rate} = \frac{\Delta E_j}{-\Delta E_i}$$

Benchmarks:

- 0: no foreign offset
- Between 0 and 1: global emissions still fall, but less than the domestic reduction
- 1: one-for-one offset, as in this small-open-economy example with identical intensities
- Greater than 1: possible if output shifts to dirtier foreign producers

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Leakage is **larger** when the world price moves less in response to the tax:

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- With little price movement, domestic demand doesn't change and imports replace the lost domestic output

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Leakage is **larger** when foreign replacement is elastic or dirty:

- Foreign supply is elastic and trade costs are low, so firms abroad can expand output quickly when home production falls
- Foreign production is more emissions-intensive than home production ($e_j > e_i$), so each unit shifted abroad offsets more of the home emissions reduction

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- The taxing country is large in world markets, so its supply cut changes the world price
- With price increases, domestic quantity demanded falls
- Then fewer imports are needed to fill the gap, so there is less of an emissions increase abroad, and greater global emissions reductions

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Leakage is **smaller** when foreign expansion is constrained or cleaner:

- If foreign producers supply is inelastic, firms abroad are limited in how much they can increase production
- If home production is dirtier than foreign production ($e_i > e_j$), shifting output abroad offsets less of the domestic abatement and can even lower global emissions

Carbon tax + carbon tariff

Adding a carbon tariff

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Here we will tax imports at the same rate as domestic production:

$$\tau = t = e \times d$$

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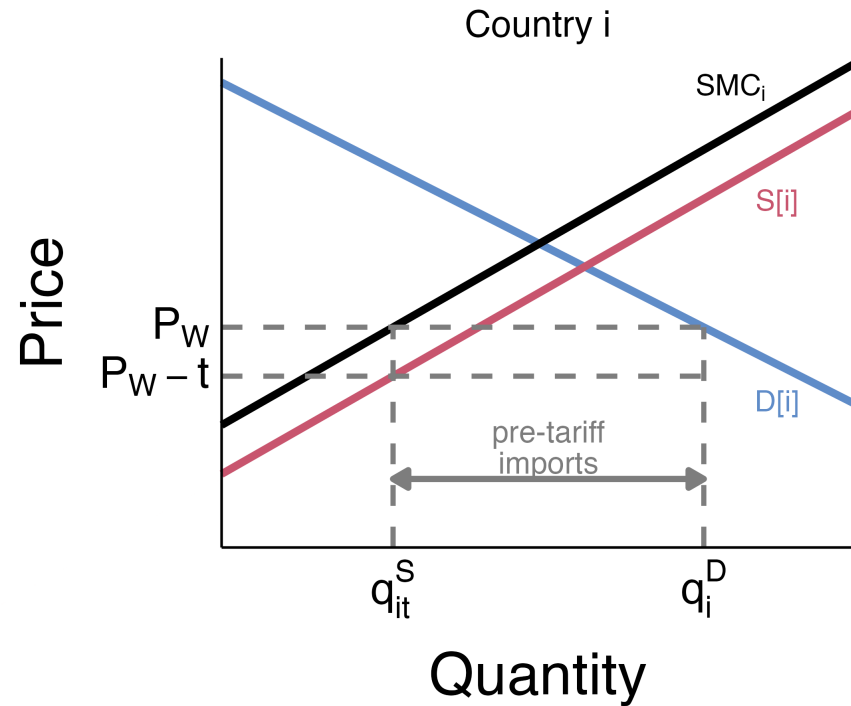
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A carbon tariff raises the price producers receive by $\tau = t$, so producers get $P_W - t + \tau = P_W$, the price without a tax or tariff

A carbon tariff raises the price consumers pay by $\tau = t$, so consumers pay $P_W + \tau$

So tax + tariff works like a consumption tax in the end

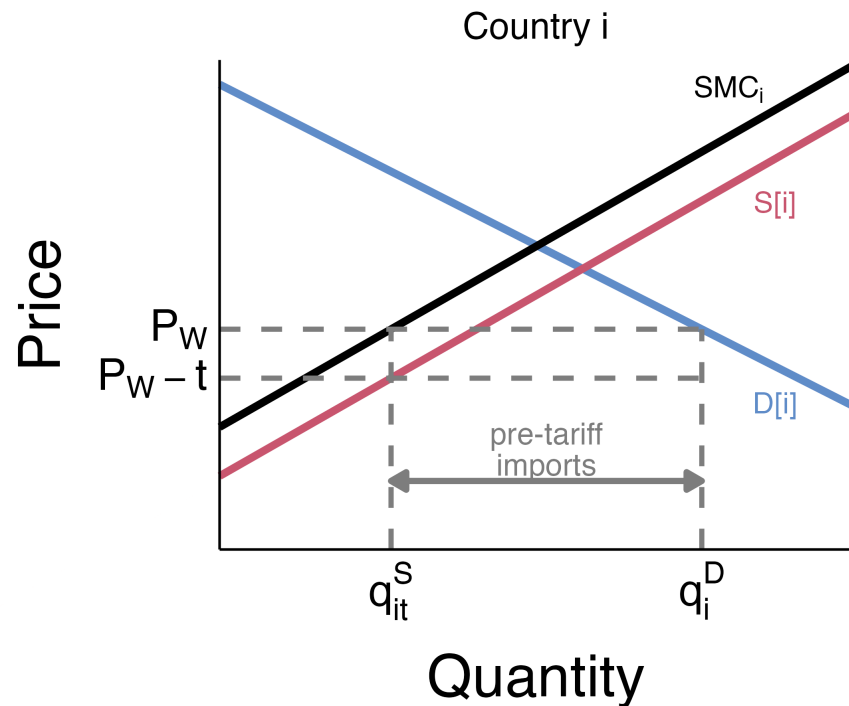
Adding a carbon tariff



Start from the carbon tax alone:

- Producers net $P_W - t$
- Home output is q_{it}^S
- Consumers still pay P_W and buy q_i^D
- Country i imports $q_i^D - q_{it}^S$

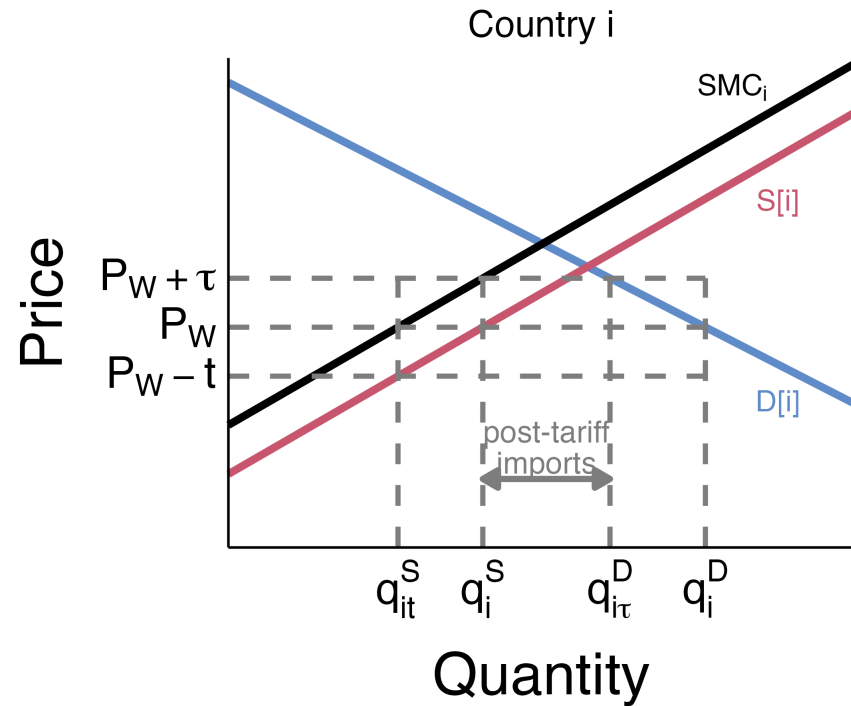
Adding a carbon tariff



Why is leakage the problem here?

- Imports replace the lost domestic output
- So home emissions fall, but foreign production rises
- The tax changes domestic supply, not consumption
- Since consumption doesn't change, there must be more imports

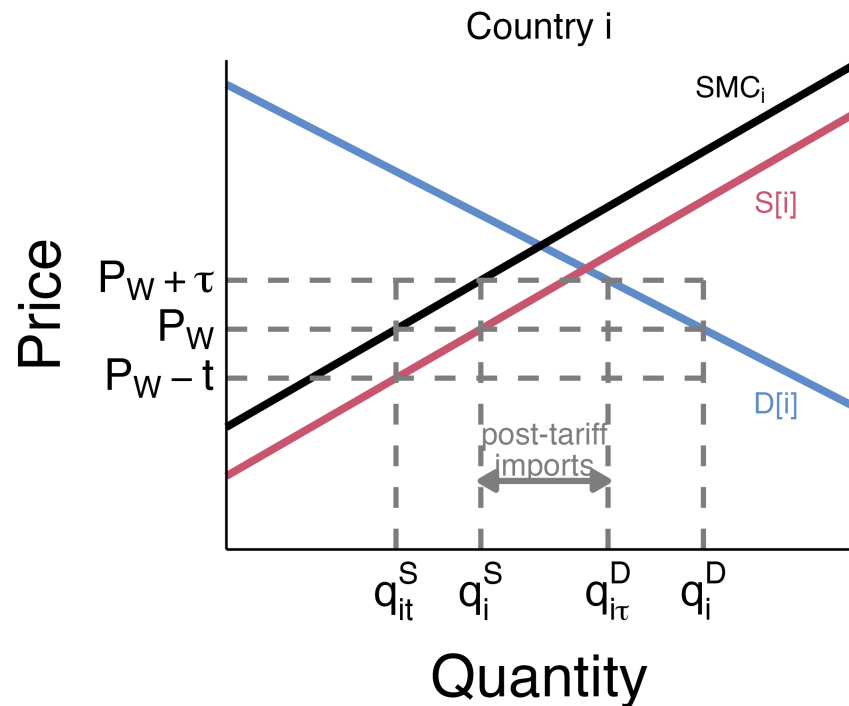
Adding a carbon tariff



Now add a carbon tariff $\tau = t$ on imports:

- Consumers pay $P_W + \tau$
- Domestic producers net $P_W + \tau - t = P_W$
- So domestic supply returns to q_i^S

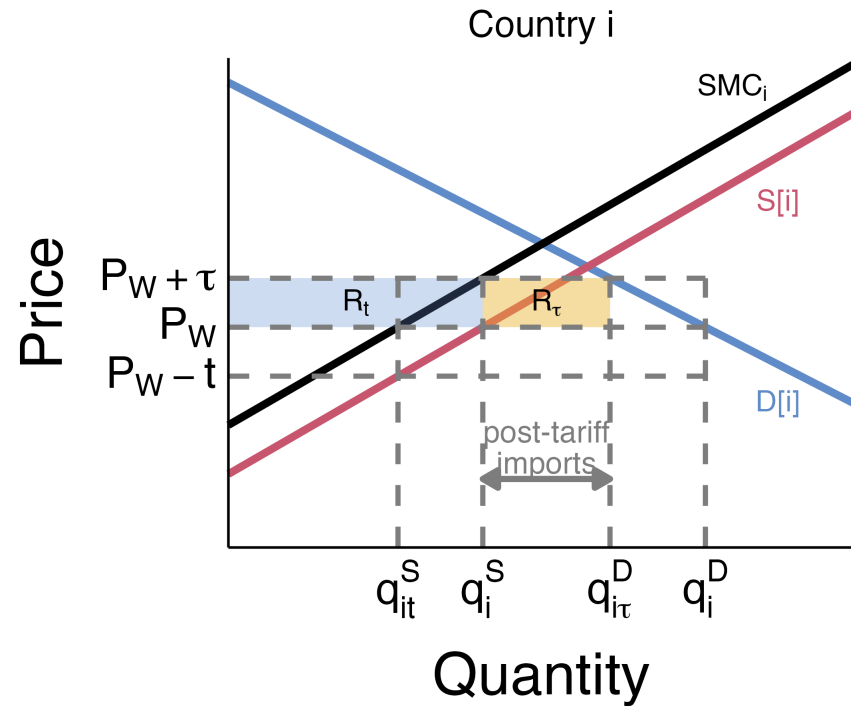
Adding a carbon tariff



Total supply and demand effects of tax+tariff:

- No net effect on producers since their quantity is the same as unregulated
- Consumers face a higher price $P_W + \tau$, reduce consumption, and reduce imports: **global production and emissions fall**

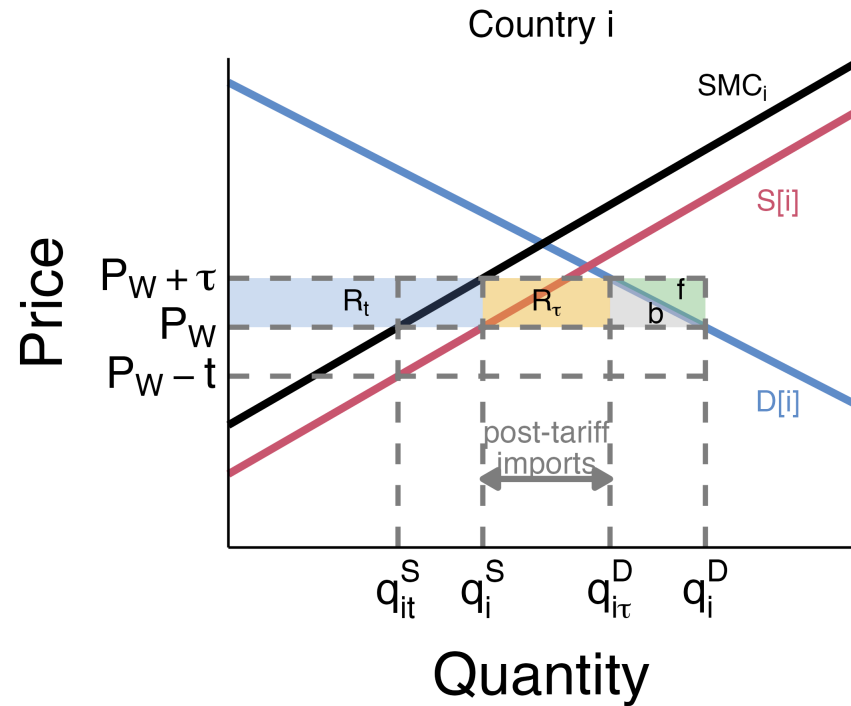
Adding a carbon tariff



Revenue effects:

- The domestic carbon tax still raises R_t
- The tariff raises R_τ on remaining imports
- Both are transfers inside country i : tax R_t paid by producers, tariff R_τ paid by consumers

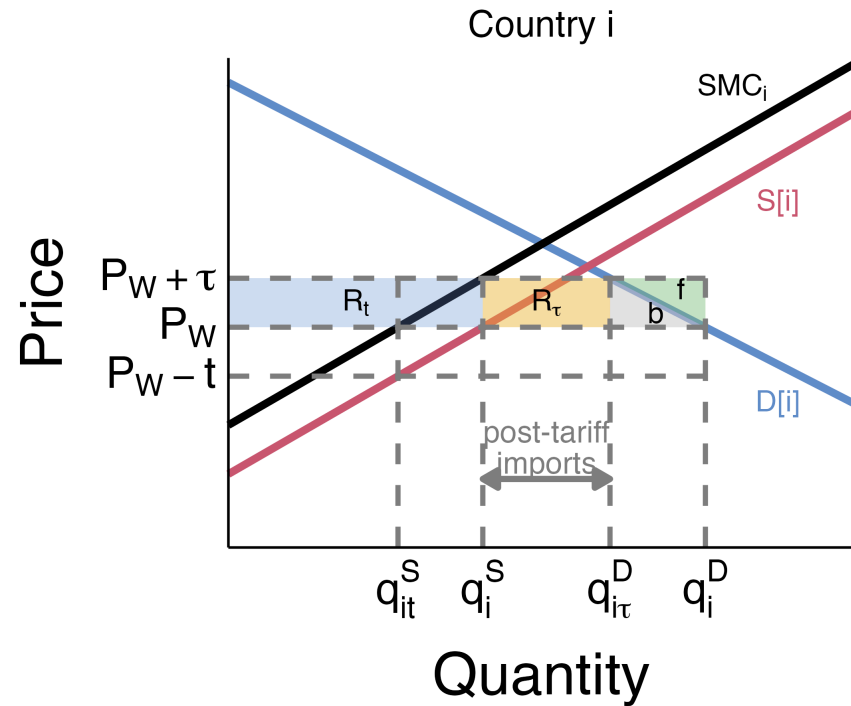
Adding a carbon tariff



Welfare and environment:

- b is the consumption deadweight loss from the higher price
- Reduced consumption where the private benefit $D[i]$ was greater than the PMC of production P_W

Adding a carbon tariff



Welfare and environment:

- $b + f$ is the environmental gain from lower world output and emissions of $q_i^D - q_{i\tau}^D$
- Tax + tariff improves social welfare by f

Recap

With only a tax:

- Domestic production/emissions become more expensive
- Consumers can easily switch to imports: "leakage"
- We taxed production but received no environmental benefits

Adding a carbon tariff:

- We are now making foreign production/emissions more expensive for our domestic consumers
- Consumers can no longer as easily switch to imports
- Emissions fall because they are correctly priced (good!)
- Consumption falls (bad!)

The key insight

With identical carbon intensities everywhere:

The key insight

With identical carbon intensities everywhere:

Carbon tax + carbon tariff = Consumption tax

The key insight

With identical carbon intensities everywhere:

Carbon tax + carbon tariff = Consumption tax

Why?

- Tax on domestic production raises domestic costs
- Tariff on imports raises import costs
- Both domestic and imported goods face the same carbon price
- Equivalent to taxing consumption of the carbon-intensive good

Numerical example: setup

Use the same simple market for both equivalence results:

$$Q_D = 100 - P, \quad Q_S = P - 20, \quad P_W = 40$$

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$$Q_D = 100 - P, \quad Q_S = P - 20, \quad P_W = 40$$

- Q_D = domestic quantity demanded
- Q_S = domestic quantity supplied
- $M = Q_D - Q_S$ = imports
- P_W = world price

Numerical example: setup

Carbon and policy terms:

- e_H and e_F = home and foreign emissions per unit
- d = damage per unit of emissions
- t = domestic carbon tax on home production
- τ = carbon tariff on imports

$$e_H = e_F = 1, \quad d = 10, \quad t = \tau = 10$$

Numerical example: setup

Start from the free-trade benchmark:

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Step 1: under free trade, the domestic price equals the world price

$$P = P_W = 40$$

Numerical example: setup

Start from the free-trade benchmark:

Step 1: under free trade, the domestic price equals the world price

$$P = P_W = 40$$

Step 2: plug P_W into demand and supply

$$Q_D = 100 - 40 = 60$$

$$Q_S = 40 - 20 = 20$$

$$M = Q_D - Q_S = 60 - 20 = 40$$

Numerical example: tax + tariff

Under a domestic carbon tax and equal tariff, work in three steps:

Numerical example: tax + tariff

Under a domestic carbon tax and equal tariff, work in three steps:

Step 1: solve for prices

Here P_c is the price paid by consumers and P_p is the net price received by domestic producers

$$P_c = P_W + \tau = 40 + 10 = 50$$

$$P_p = P_c - t = 50 - 10 = 40$$

Numerical example: tax + tariff

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 40 - 20 = 20$$

$$M = Q_D - Q_S = 50 - 20 = 30$$

Numerical example: tax + tariff

Step 3: compute government revenue

Here R_t is domestic carbon-tax revenue and R_τ is tariff revenue

$$R_t = t \times Q_S = 10 \times 20 = 200$$

$$R_\tau = \tau \times M = 10 \times 30 = 300$$

$$R = R_t + R_\tau = 200 + 300 = 500$$

Numerical example: equivalent consumption tax

Now replace tax + tariff with a pure consumption tax of 10:

Numerical example: equivalent consumption tax

Now replace tax + tariff with a pure consumption tax of 10:

Step 1: solve for prices

Consumers pay the tax, so P_c rises, but producers still receive the world price

$$P_c = P_W + 10 = 40 + 10 = 50$$

$$P_p = P_W = 40$$

Numerical example: equivalent consumption tax

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 40 - 20 = 20$$

$$M = Q_D - Q_S = 50 - 20 = 30$$

Numerical example: equivalent consumption tax

Step 3: compute government revenue

$$R_c = 10 \times Q_D = 10 \times 50 = 500$$

Numerical example: equivalent consumption tax

Step 3: compute government revenue

$$R_c = 10 \times Q_D = 10 \times 50 = 500$$

So tax + tariff and a consumption tax generate the same outcome:

- same consumer price: $P_c = 50$
- same producer price: $P_p = 40$
- same domestic demand: $Q_D = 50$
- same domestic supply: $Q_S = 20$
- same imports: $M = 30$
- same total revenue: 500

Carbon tariff alone

What if we only use a tariff?

Some proposals call for carbon tariffs **without** a domestic carbon tax

What if we only use a tariff?

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What does this accomplish?

What if we only use a tariff?

Key intuition:

A carbon tariff raises the price producers receive by τ , which is the same thing as a production subsidy of τ

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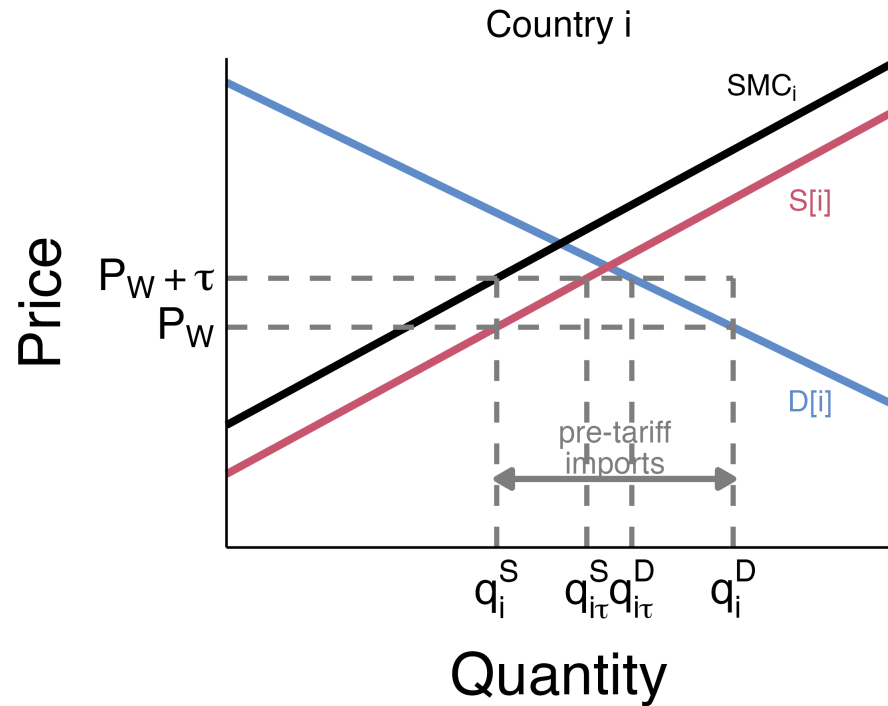
A carbon tariff raises the price consumers pay by τ , which is the same thing as a consumption tax of τ

So tariff alone works like a consumption tax plus a production subsidy in the end

Carbon tariff alone in practice



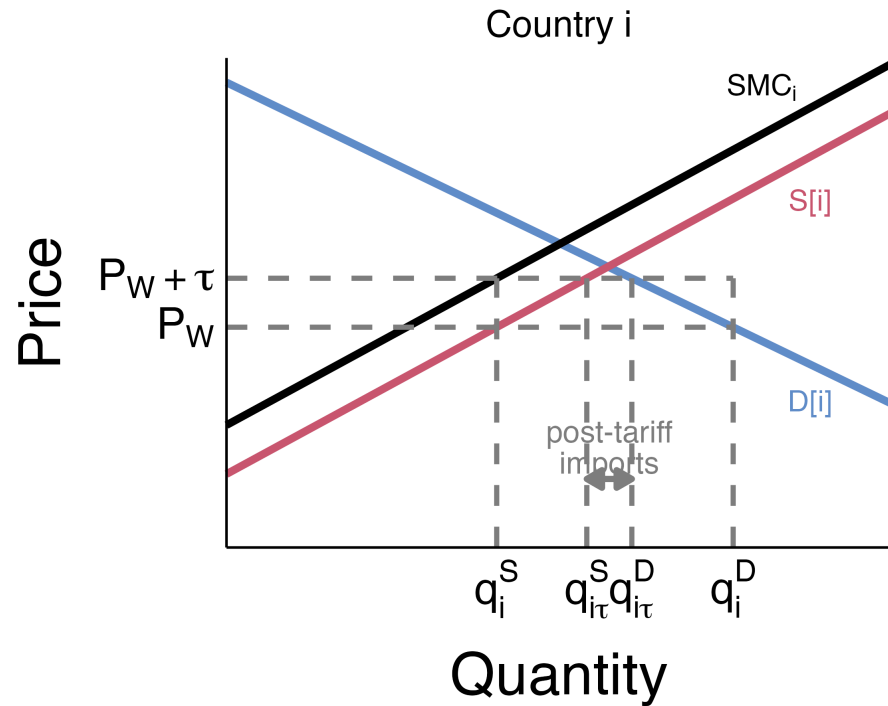
Carbon tariff alone



Start from free trade:

- Consumers and producers in country i both face P_W
- Domestic output is q_i^S and consumption is q_i^D
- Imports fill the gap $q_i^D - q_i^S$

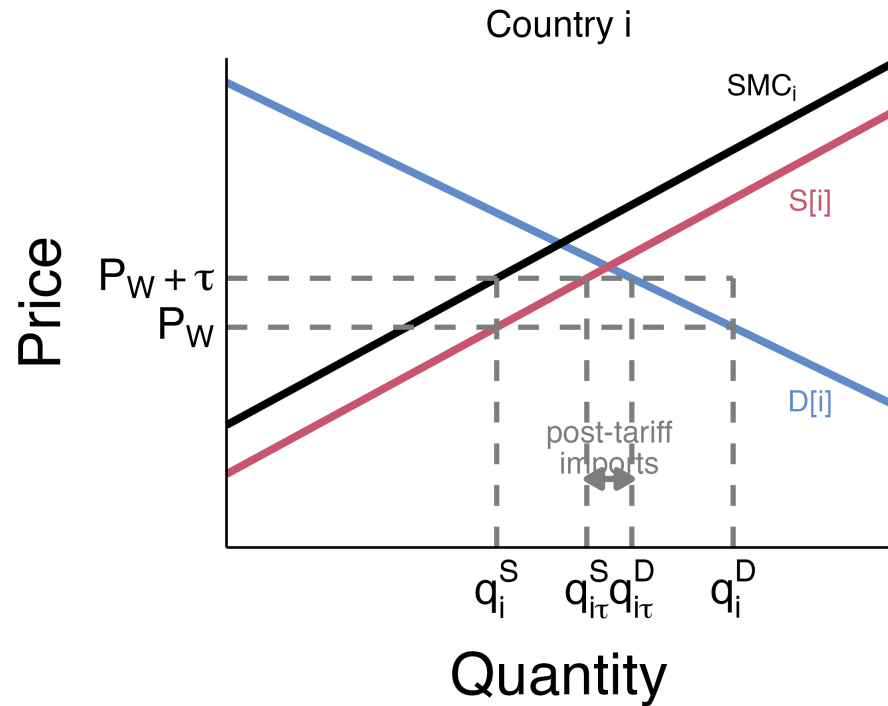
Carbon tariff alone



Now impose a carbon tariff τ with no domestic carbon tax:

- Consumers pay $P_W + \tau$
- Domestic producers also receive $P_W + \tau$
- So consumption falls and home output rises

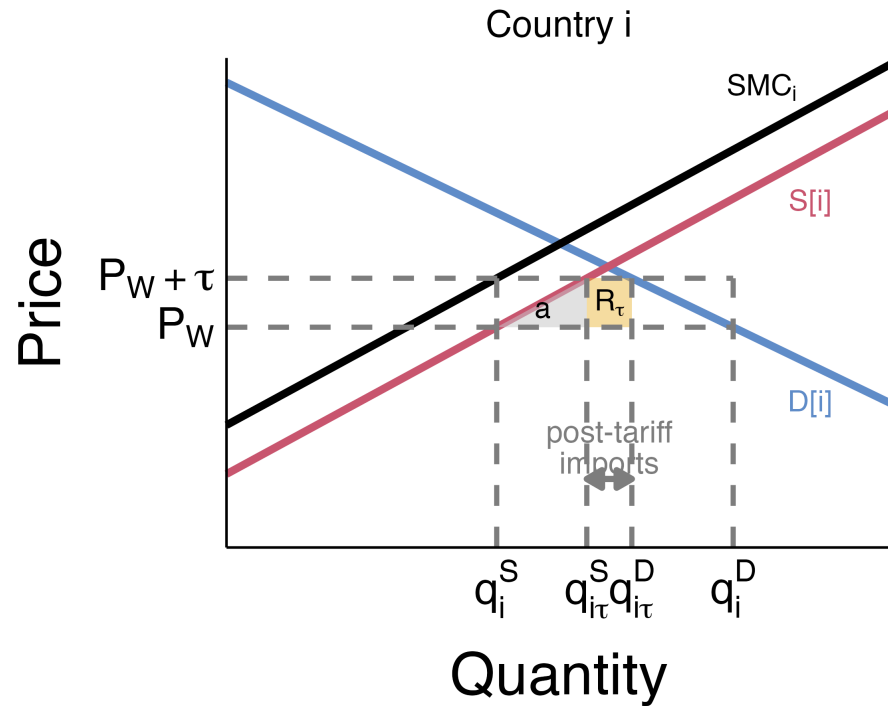
Carbon tariff alone



Why is this different from tax + tariff?

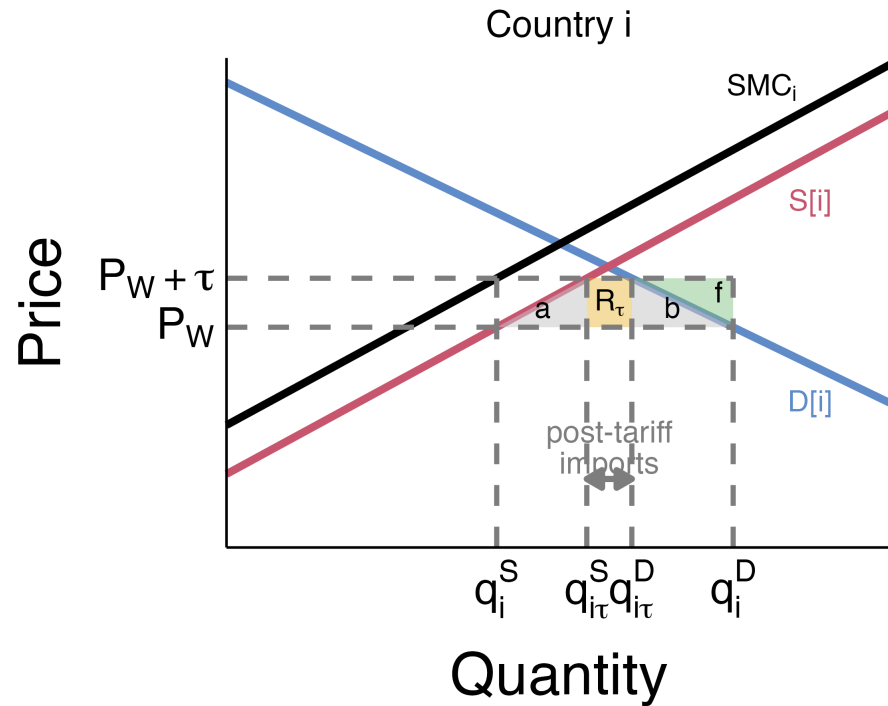
- The tariff alone taxes consumption but also protects domestic producers
- Output rises to $q_{i\tau}^S$ and demand falls to $q_{i\tau}^D$
- Imports shrink because both margins move

Carbon tariff alone



- The government collects tariff revenue R_τ on remaining imports
- The tariff pushes some production away from cheaper foreign suppliers

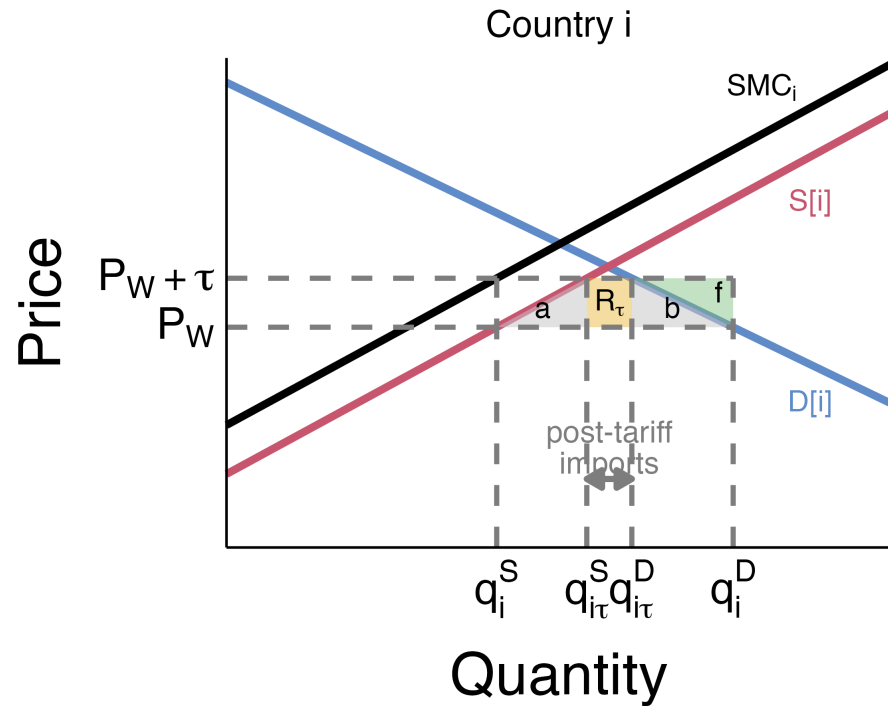
Carbon tariff alone



- a is the production DWL from replacing cheaper imports with costlier home output
- b is the consumption DWL from reduced consumption at the higher price
- Lower consumption has environmental benefits $b + f$:

$$(q_i^D - q_{i\tau}^D) \times d$$

Carbon tariff alone



Welfare and environment:

- Net social benefit is $(b + f) - (b + a) = f - a$
- Gains from real reductions in pollution
- Losses from inefficient subsidization of domestic production
- Net effect is ambiguous

Numerical example: setup

For the tariff-alone example, use the same market structure:

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- P_p = price received by domestic producers

Policy for this case:

$$t = 0, \quad \tau = 10$$

Numerical example: setup

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Free-trade benchmark:

$$P_c = P_p = P_W = 40$$

$$Q_D = 100 - 40 = 60$$

$$Q_S = 40 - 20 = 20$$

$$M = Q_D - Q_S = 40$$

Numerical example: tariff alone

With a tariff alone, work in three steps:

Numerical example: tariff alone

With a tariff alone, work in three steps:

Step 1: solve for prices

Here both consumers and domestic producers face the tariff-inclusive price

$$P_c = P_W + \tau = 40 + 10 = 50$$

$$P_p = P_c = 50$$

Numerical example: tariff alone

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 50 - 20 = 30$$

$$M = Q_D - Q_S = 50 - 30 = 20$$

Numerical example: tariff alone

Step 3: compute tariff revenue

$$R_{\tau} = \tau \times M = 10 \times 20 = 200$$

Numerical example: equivalent tax + subsidy

Now replace the tariff with a consumption tax of 10 and a production subsidy of 10:

Numerical example: equivalent tax + subsidy

Now replace the tariff with a consumption tax of 10 and a production subsidy of 10:

Step 1: solve for prices

Consumers pay the consumption tax, and producers receive the world price plus the subsidy

$$P_c = P_W + 10 = 40 + 10 = 50$$

$$P_p = P_W + 10 = 40 + 10 = 50$$

Numerical example: equivalent tax + subsidy

Step 2: plug those prices into demand and supply

$$Q_D = 100 - P_c = 100 - 50 = 50$$

$$Q_S = P_p - 20 = 50 - 20 = 30$$

$$M = Q_D - Q_S = 50 - 30 = 20$$

Numerical example: equivalent tax + subsidy

Step 3: compute the government budget

Here R_c is consumption-tax revenue

$$R_c = 10 \times Q_D = 10 \times 50 = 500$$

$$\text{Subsidy cost} = 10 \times Q_S = 10 \times 30 = 300$$

$$R^{net} = R_c - \text{Subsidy cost} = 500 - 300 = 200$$

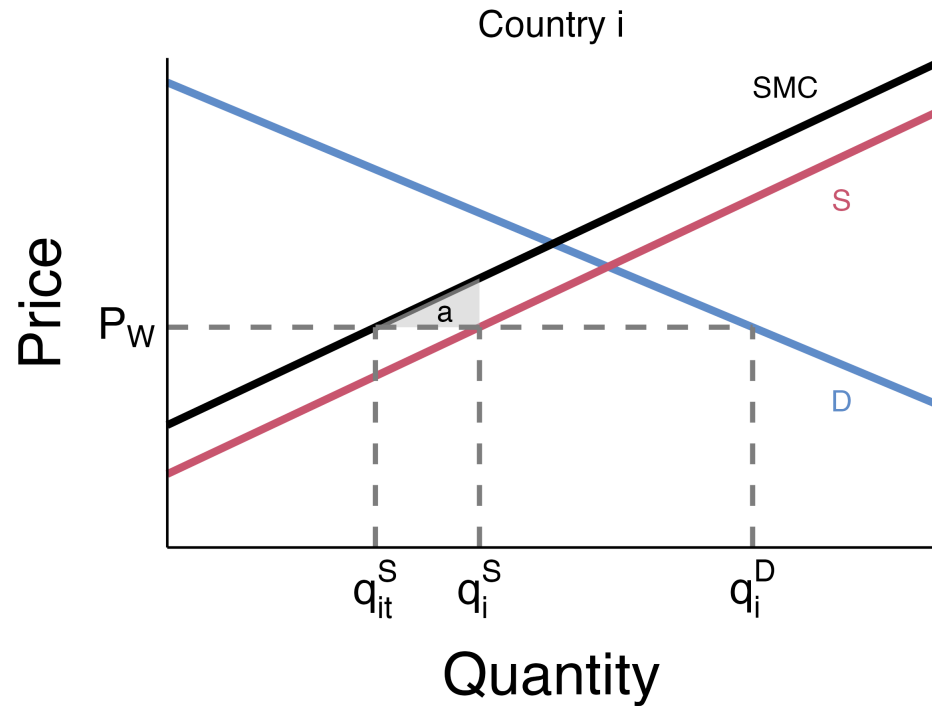
Numerical example: equivalent tax + subsidy

So tariff alone and a consumption tax plus production subsidy generate the same outcome:

- same consumer price: $P_c = 50$
- same producer price: $P_p = 50$
- same domestic demand: $Q_D = 50$
- same domestic supply: $Q_S = 30$
- same imports: $M = 20$
- same net government revenue: 200

World carbon tax

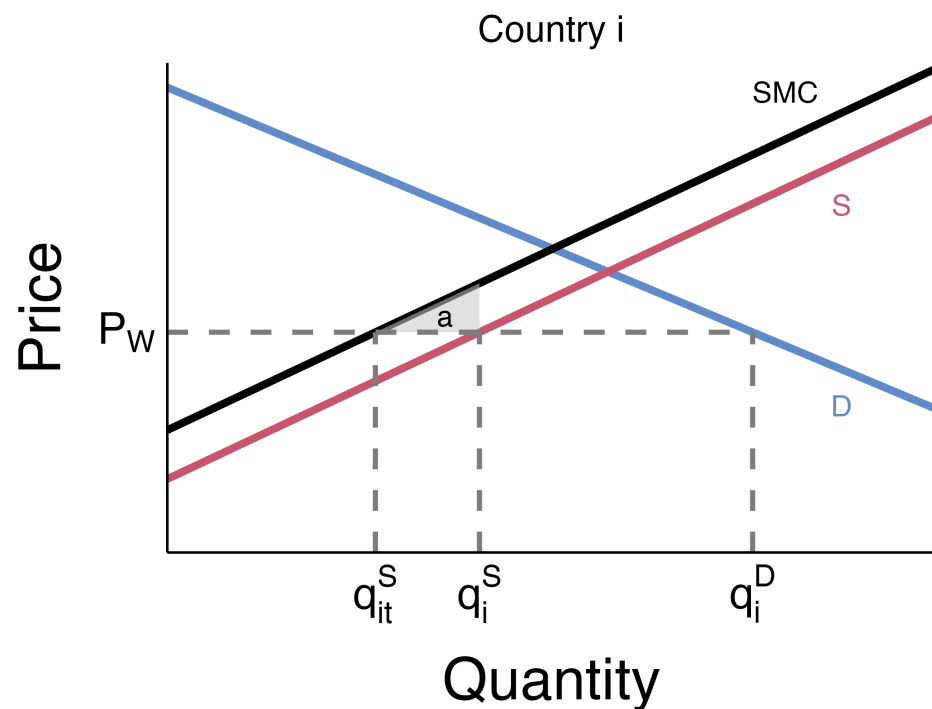
World carbon tax



Suppose the rest of the world already prices carbon so P_W reflects the carbon cost *abroad*:

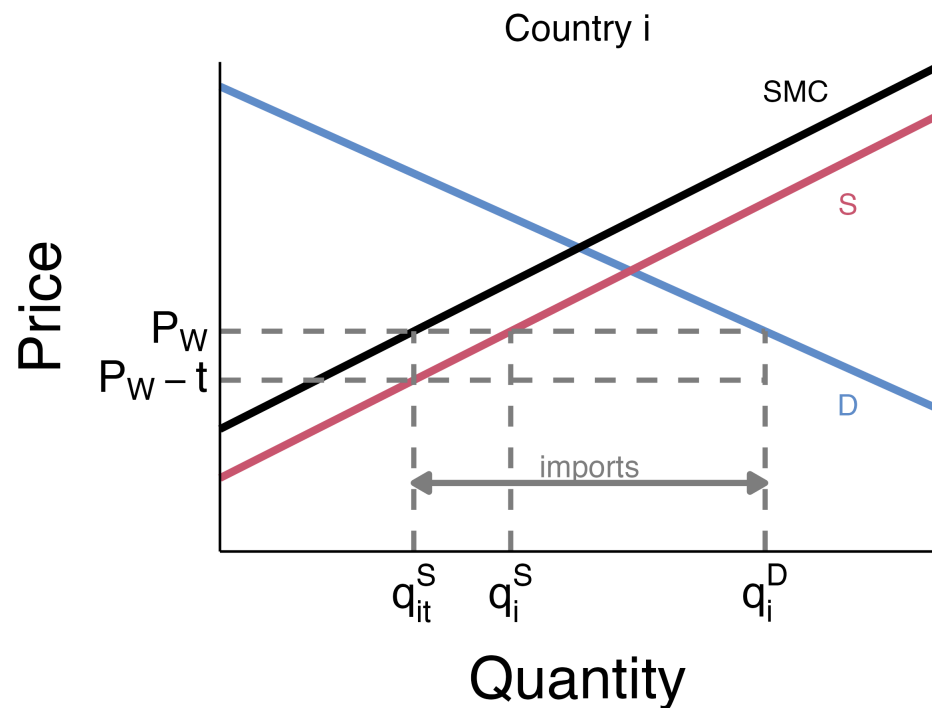
- Country i still produces q_{it}^S without its own carbon tax
- The efficient output for country i is q_i^S , where $SMC = P_W$

World carbon tax



- At q_i^S , SMC exceeds the revenue from selling the good at P_W
- $q_i^S - q_{it}^S$ is leakage *into* the country from carbon taxation abroad
- a is the resulting deadweight loss from overproduction domestically

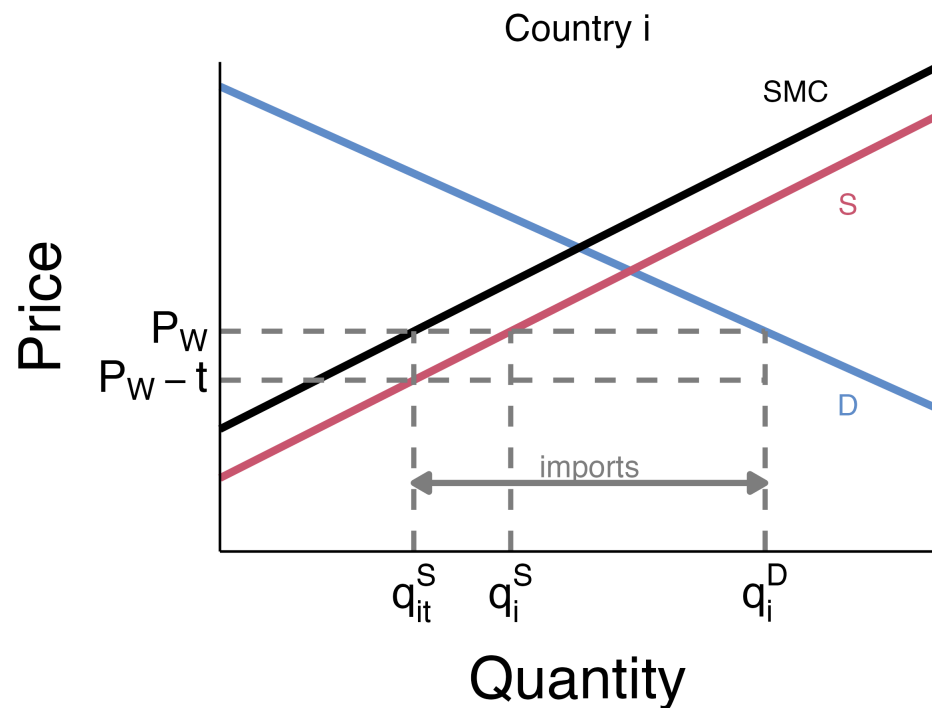
World carbon tax



If country i also imposes the carbon tax t :

- Producers receive net-of-tax price $P_W - t$
- Supply falls from q_i^S to q_{it}^S
- Imports rise to $q_i^D - q_{it}^S$
- Consumers still face the world price P_W

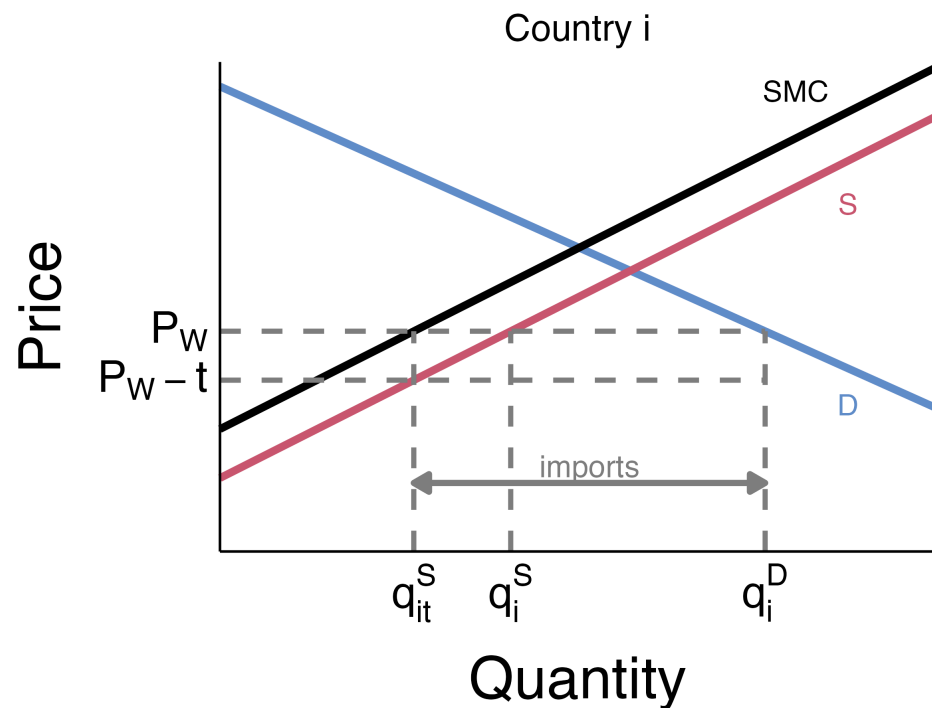
World carbon tax



Why are those extra imports not leakage?

- Foreign producers already face the same carbon price
- The earlier distortion was leakage from the rest of the world to country i
- Those imports reverse that leakage
- The DWL a disappears

World carbon tax



At that point the allocation is efficient:

- Every country has internalized the externality
- Global SMC equals the marginal benefit of the last unit consumed
- Production is cost-effective across countries
- A carbon tariff is unnecessary

Complications

Real-world complications

Several factors complicate carbon tariff design:

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1. **Resource shuffling:** clean production exported, dirty production consumed domestically

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Real-world complications

Several factors complicate carbon tariff design:

1. **Resource shuffling:** clean production exported, dirty production consumed domestically
2. **Non-price policies:** regulations vs. carbon prices
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4. **Retaliation:** trading partners may impose counter-tariffs

Retaliation risk

Trading partners may retaliate



Resource shuffling

Suppose country j has both clean and dirty production

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With a carbon tariff based on average intensity:

- Export the clean stuff to country i
- Consume the dirty stuff domestically

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Resource shuffling

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This is **another form of leakage**

Non-price policies

Many countries use regulations rather than carbon prices:

- Fuel economy standards
- Renewable portfolio standards
- Building codes

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How do you calculate the "equivalent carbon price"?

Non-price policies

Many countries use regulations rather than carbon prices:

- Fuel economy standards
- Renewable portfolio standards
- Building codes

How do you calculate the "equivalent carbon price"?

Very difficult: may require complex modeling

Export rebates

Should country i rebate carbon taxes on exports?

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Arguments for:

- Maintains competitiveness
- Symmetric with import tariffs

Export rebates

Should country i rebate carbon taxes on exports?

Arguments for:

- Maintains competitiveness
- Symmetric with import tariffs

Arguments against:

- Reduces domestic incentive to decarbonize
- May look like an export subsidy (WTO issues)

WTO compatibility

Carbon tariffs face WTO scrutiny:

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EU CBAM designed carefully to fit within WTO rules

Key takeaways
class: lecture, venue, middle name: takeaways

Summary

1. **Carbon tariffs** aim to address leakage and competitiveness concerns

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Summary

1. **Carbon tariffs** aim to address leakage and competitiveness concerns
2. **Carbon tax + tariff** is equivalent to a consumption tax
 - Reduces consumption of carbon-intensive goods
 - Can improve welfare if environmental benefit exceeds distortion costs
3. **Carbon tariff alone** = consumption tax + production subsidy
 - May increase domestic emissions
 - Less effective than tax + tariff

Summary (continued)

1. **World carbon tax** is first-best
 - Carbon tariffs are second-best when global coordination fails

Summary (continued)

1. **World carbon tax** is first-best

- Carbon tariffs are second-best when global coordination fails

2. **Different carbon intensities** create measurement challenges

- Setting tariff = foreign intensity is efficient but looks protectionist
- Setting tariff = domestic intensity under-prices foreign emissions

Summary (continued)

1. **World carbon tax** is first-best

- Carbon tariffs are second-best when global coordination fails

2. **Different carbon intensities** create measurement challenges

- Setting tariff = foreign intensity is efficient but looks protectionist
- Setting tariff = domestic intensity under-prices foreign emissions

3. **Many complications** in practice

- Resource shuffling, non-price policies, export rebates, WTO rules

The bottom line

Carbon tariffs are a **complement** to carbon pricing, not a substitute

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They work best when:

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- Set based on actual carbon content
- Part of broader international coordination

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They work best when:

- Combined with a domestic carbon tax
- Set based on actual carbon content
- Part of broader international coordination

Alone, they are an imperfect tool that may create new distortions